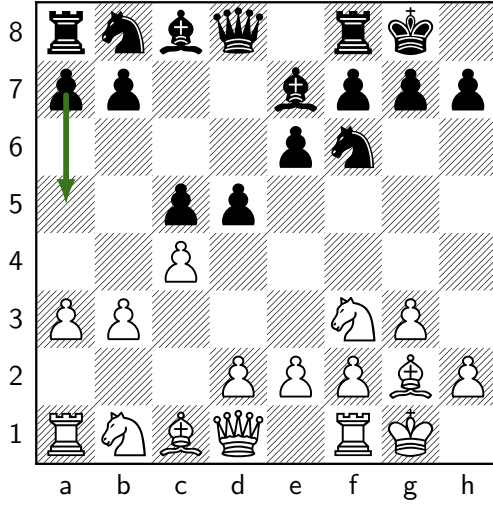


Melanie-BOTS (CCRL engines) — Reti Opening

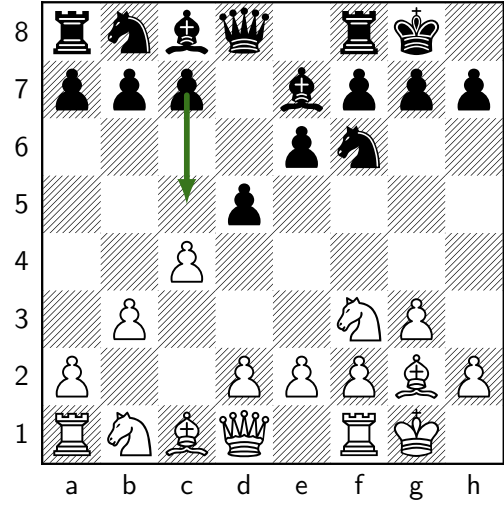
Melanie (Deep_Vision)

Abstract

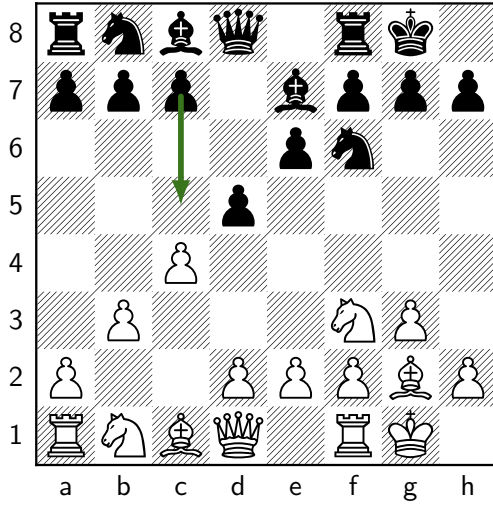
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



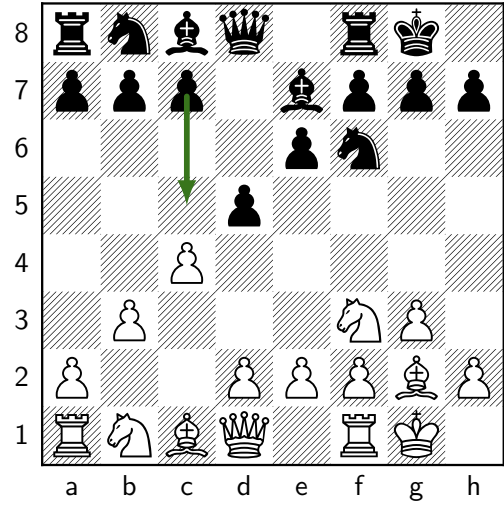
(a)



(b)

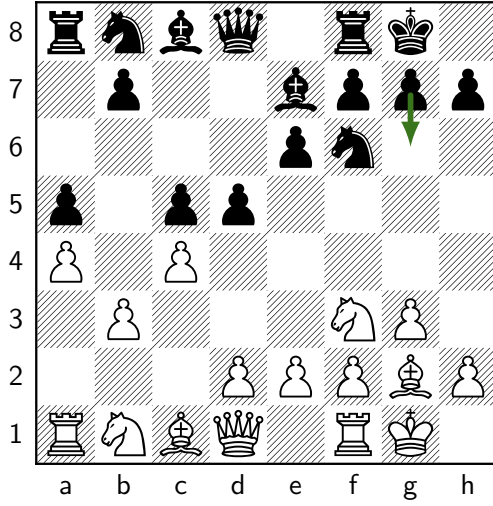


(c)

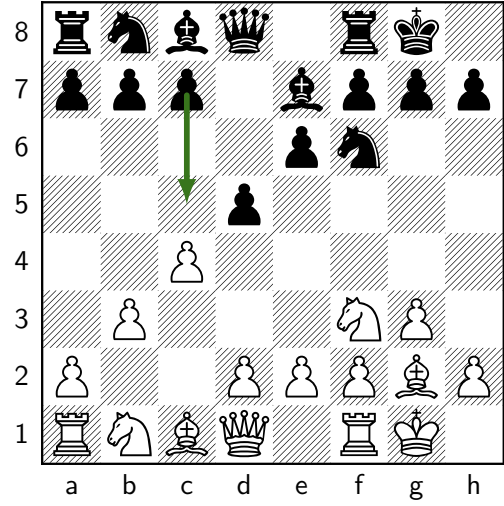


(d)

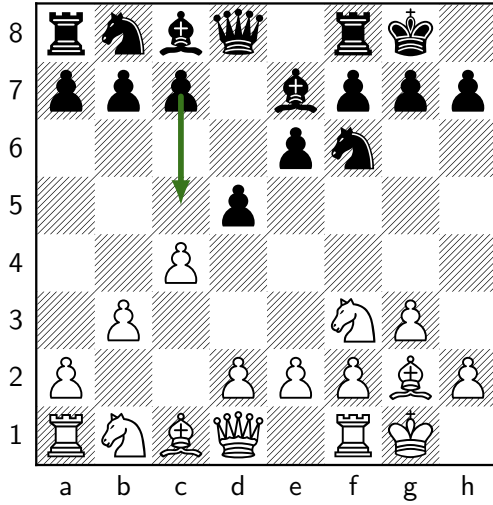
Figure 1: **Move 7, Black: a5**. (a) Our position; Melanie plays a5 (a7a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1254` (ply 11) [structural distance $d=21$, QAP obj=59]: the master played c5; under μ our *Pa2* plays the role of that game’s *Pc2*; cross-type roles: our *bg2* as their *Ra8*, our *Pb7* as their *Nb8*; transferred move a7a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1255` (ply 11) [structural distance $d=21$, QAP obj=59]: the master played c5; under μ our *Pa2* plays the role of that game’s *Pc2*; cross-type roles: our *bg2* as their *Ra8*, our *Pb7* as their *Nb8*; transferred move a7a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3784` (ply 11) [structural distance $d=21$, QAP obj=59]: the master played c5; under μ our *Pa2* plays the role of that game’s *Pc2*; cross-type roles: our *bg2* as their *Ra8*, our *Pb7* as their *Nb8*; transferred move a7a5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to a5 (summed QAP objective 471, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



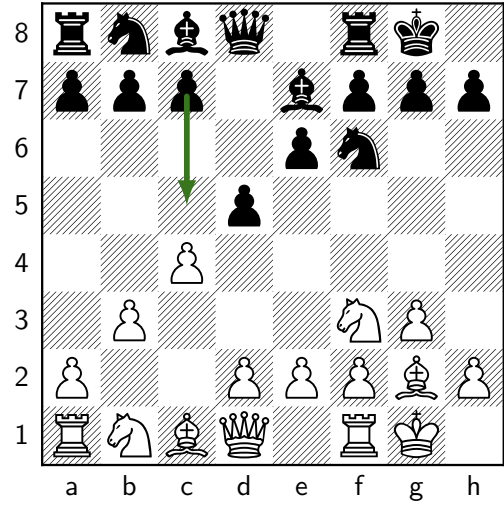
(a)



(b)

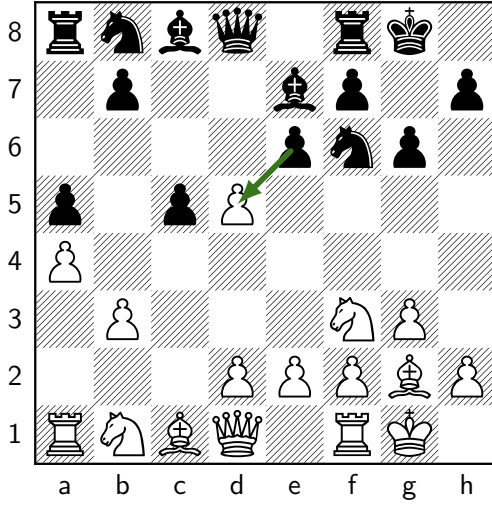


(c)

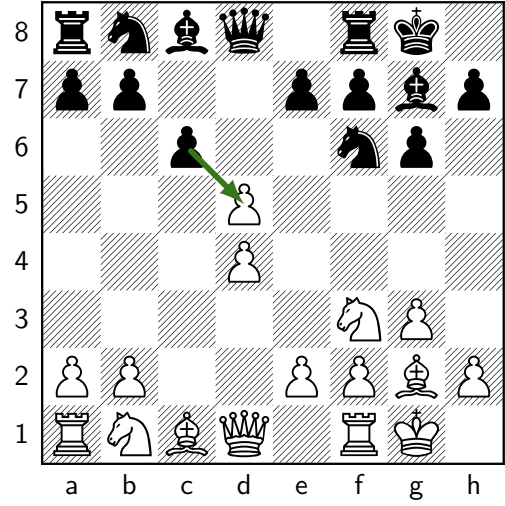


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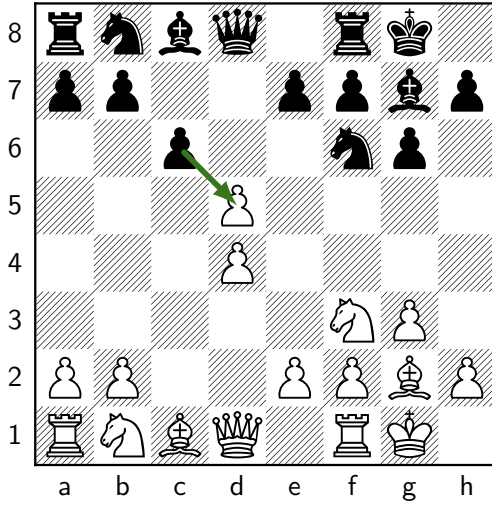
Figure 2: **Move 8, Black: g6.** (a) Our position; Melanie plays **g6** (**g7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1254` (ply 11) [structural distance $d=21$, QAP obj=56]: the master played **c5**; under μ our $Pg2$ plays the role of that game’s $Pc2$; cross-type roles: our $pg3$ as their $Ra8$, our $Qd8$ as their $Nb8$; transferred move **g7g6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1255` (ply 11) [structural distance $d=21$, QAP obj=56]: the master played **c5**; under μ our $Pg2$ plays the role of that game’s $Pc2$; cross-type roles: our $pg3$ as their $Ra8$, our $Qd8$ as their $Nb8$; transferred move **g7g6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3784` (ply 11) [structural distance $d=21$, QAP obj=56]: the master played **c5**; under μ our $Pg2$ plays the role of that game’s $Pc2$; cross-type roles: our $pg3$ as their $Ra8$, our $Qd8$ as their $Nb8$; transferred move **g7g6** (**endorses this move**). Synthesis: 8 retrieved analogs transferred to **g6** (summed QAP objective 449, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



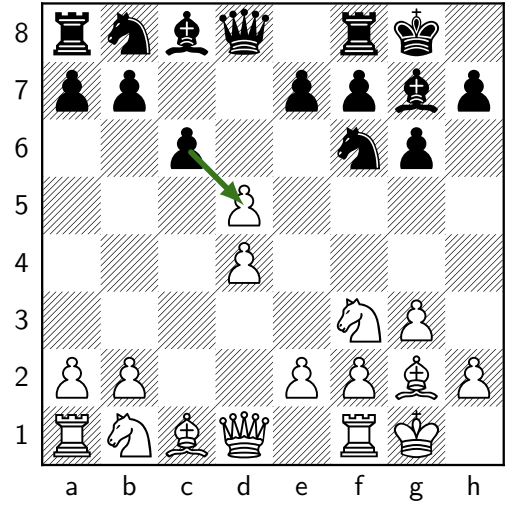
(a)



(b)

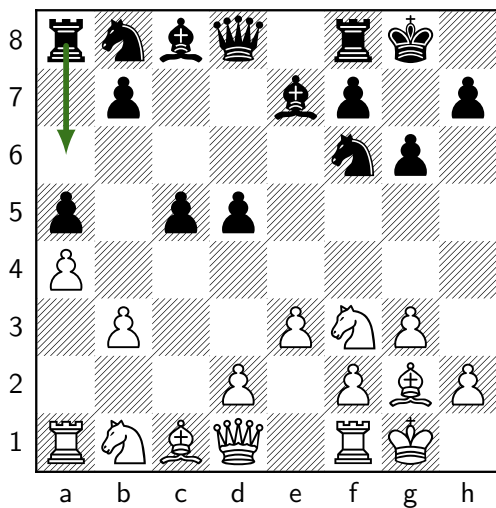


(c)

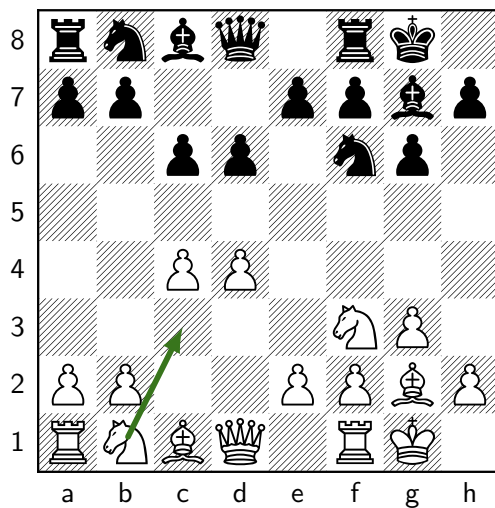


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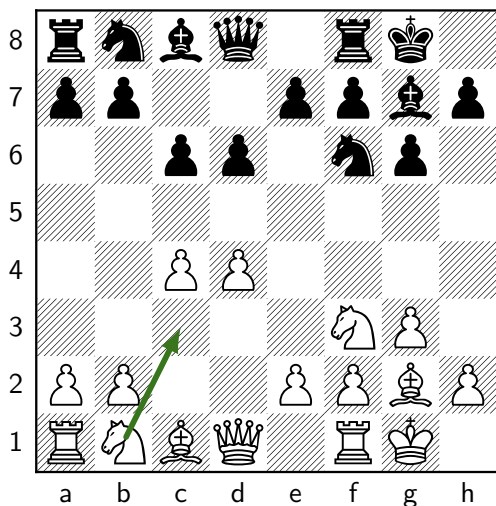
Figure 3: **Move 9, Black: exd5.** (a) Our position; Melanie plays exd5 (e6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#276 (ply 13) [structural distance $d=52$, QAP obj=94]: the master played cxd5; under μ our $Pe3$ plays the role of that game’s $Pc3$; cross-type roles: our $Pg6$ as their $Ra8$, our $pd2$ as their $Bc8$; transferred move e6d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#277 (ply 13) [structural distance $d=52$, QAP obj=94]: the master played cxd5; under μ our $Pe3$ plays the role of that game’s $Pc3$; cross-type roles: our $Pg6$ as their $Ra8$, our $pd2$ as their $Bc8$; transferred move e6d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1631 (ply 13) [structural distance $d=52$, QAP obj=94]: the master played cxd5; under μ our $Pe3$ plays the role of that game’s $Pc3$; cross-type roles: our $Pg6$ as their $Ra8$, our $pd2$ as their $Bc8$; transferred move e6d5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to exd5 (summed QAP objective 661, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



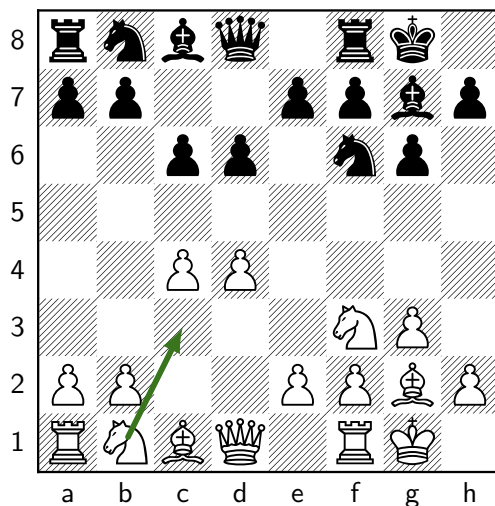
(a)



(b)

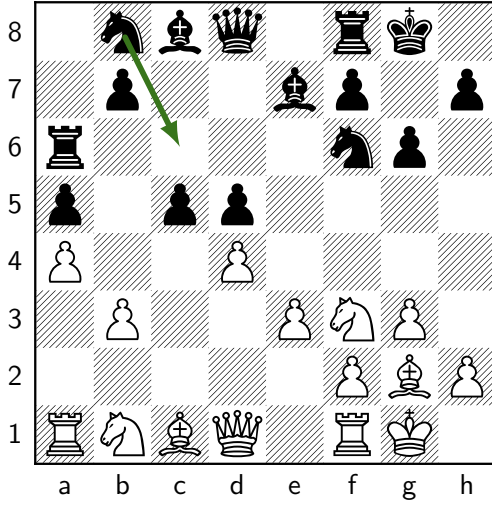


(c)

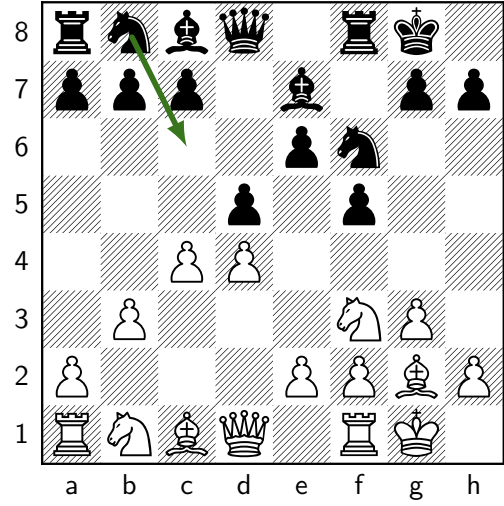


(d)

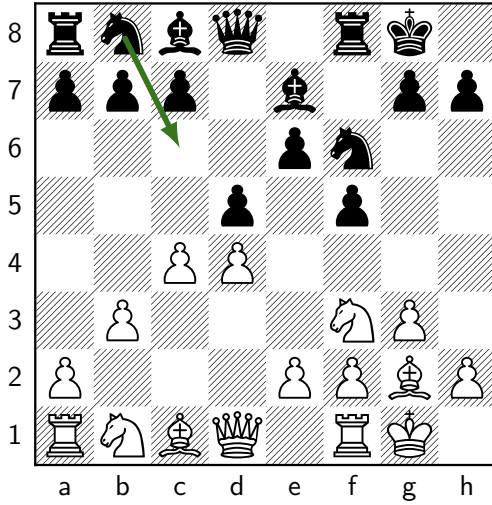
Figure 4: **Move 10, Black: Ra6.** (a) Our position; Melanie plays Ra6 (a8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#139` (ply 12) [structural distance $d=38$, QAP obj=8]: the master played Nc3; under μ our *Ra1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pe3* as their *Ra1*, our *Be7* as their *Pb2*; transferred move a8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1671` (ply 12) [structural distance $d=38$, QAP obj=8]: the master played Nc3; under μ our *Ra1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pe3* as their *Ra1*, our *Be7* as their *Pb2*; transferred move a8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1672` (ply 12) [structural distance $d=38$, QAP obj=8]: the master played Nc3; under μ our *Ra1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pe3* as their *Ra1*, our *Be7* as their *Pb2*; transferred move a8a6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Ra6 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



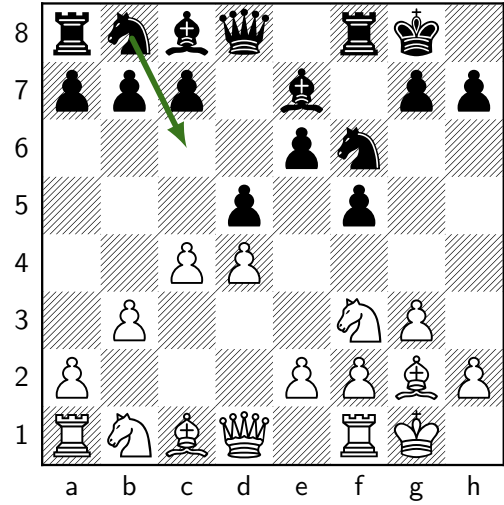
(a)



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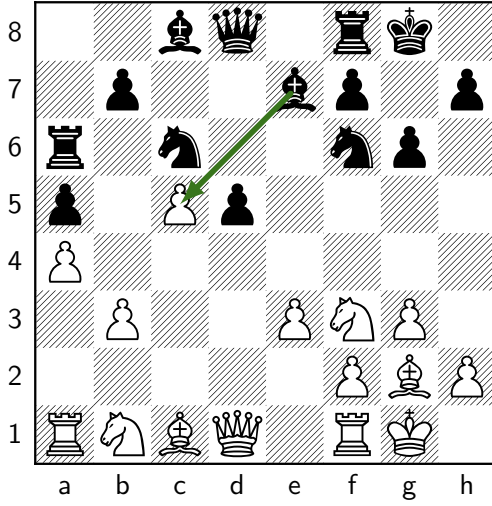


(c)

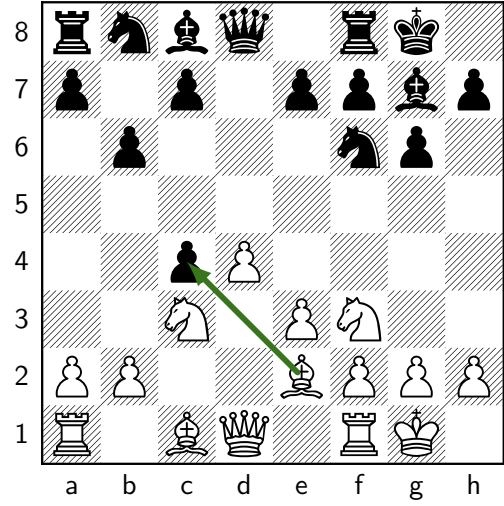


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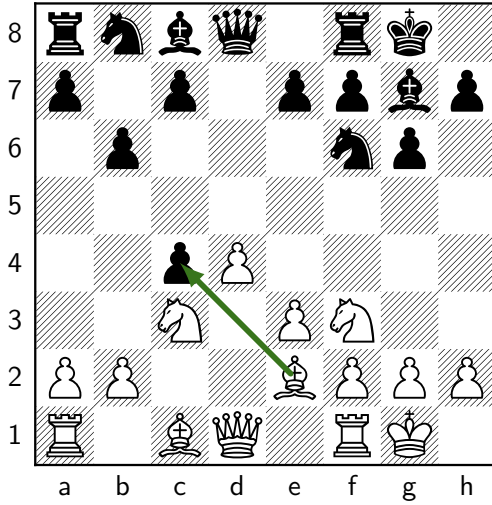
Figure 5: **Move 11, Black: Nc6.** (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#982` (ply 13) [structural distance $d=34$, QAP obj=50]: the master played Nc6; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Ph7 as their Bc8, our Be7 as their Pa7; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#983` (ply 13) [structural distance $d=34$, QAP obj=50]: the master played Nc6; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Ph7 as their Bc8, our Be7 as their Pa7; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#58` (ply 13) [structural distance $d=34$, QAP obj=50]: the master played Nc6; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Ph7 as their Bc8, our Be7 as their Pa7; transferred move b8c6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nc6 (summed QAP objective 298, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



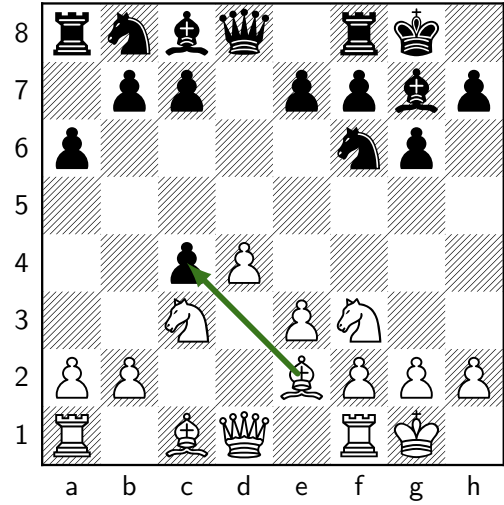
(a)



(b)

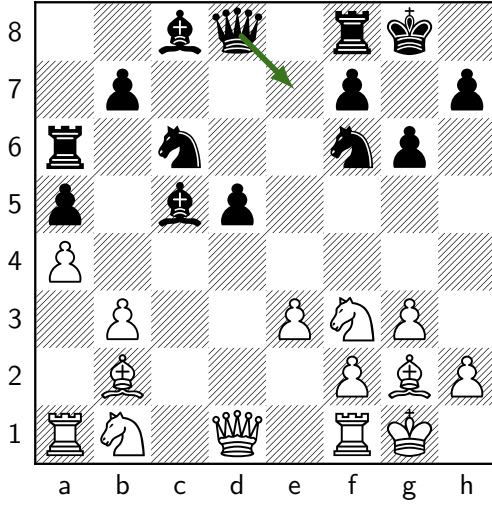


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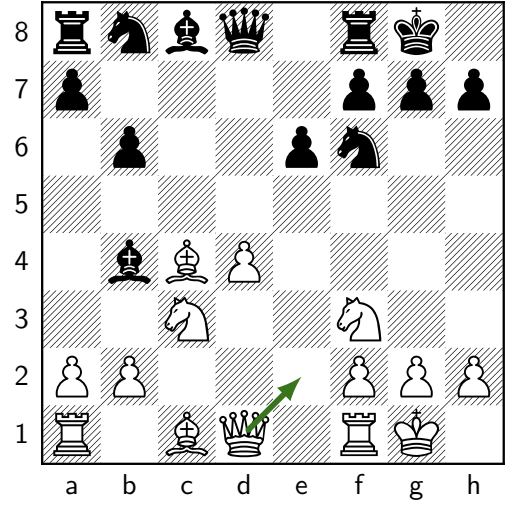


(d)

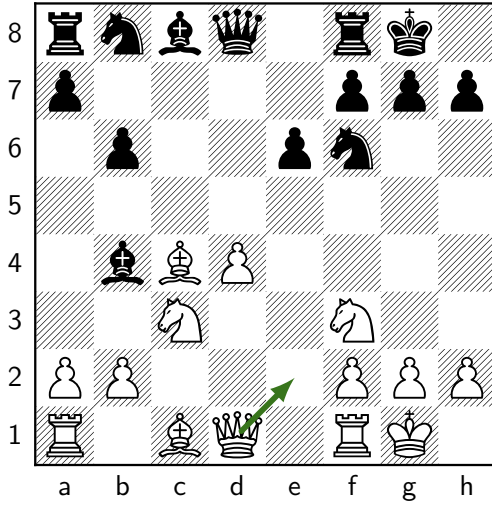
Figure 6: **Move 12, Black: Bxc5.** (a) Our position; Melanie plays Bxc5 (e7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2884` (ply 14) [structural distance $d=46$, QAP obj=77]: the master played Bxc4; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pb3* as their *Ra1*, our *pf2* as their *Bc1*; transferred move e7c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2885` (ply 14) [structural distance $d=46$, QAP obj=77]: the master played Bxc4; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pb3* as their *Ra1*, our *pf2* as their *Bc1*; transferred move e7c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3818` (ply 14) [structural distance $d=56$, QAP obj=77]: the master played Bxc4; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pb3* as their *Ra1*, our *pf2* as their *Bc1*; transferred move e7c5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bxc5 (summed QAP objective 568, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



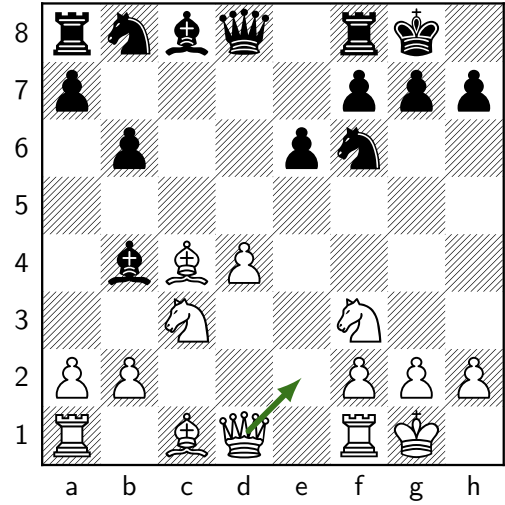
(a)



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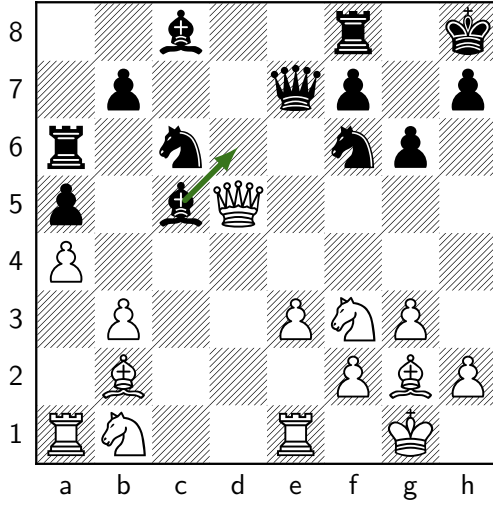


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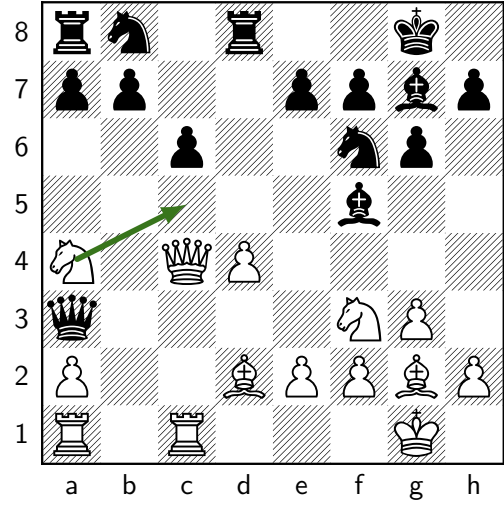


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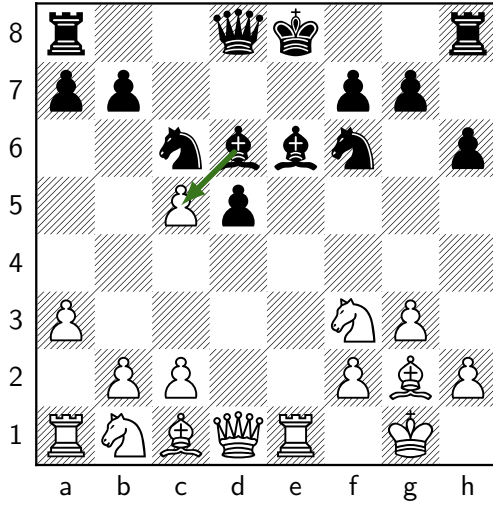
Figure 7: **Move 13, Black: Qe7.** (a) Our position; Melanie plays Qe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#712 (ply 18) [structural distance $d=48$, QAP obj=51]: the master played Qe2; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our pg3 as their Ra1, our Nc6 as their Bc1; transferred move d8e7 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#713 (ply 18) [structural distance $d=48$, QAP obj=51]: the master played Qe2; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our pg3 as their Ra1, our Nc6 as their Bc1; transferred move d8e7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1275 (ply 18) [structural distance $d=48$, QAP obj=51]: the master played Qe2; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our pg3 as their Ra1, our Nc6 as their Bc1; transferred move d8e7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qe7 (summed QAP objective 356, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



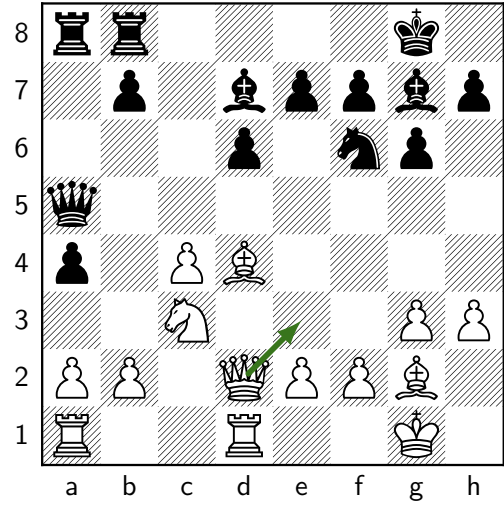
(a)



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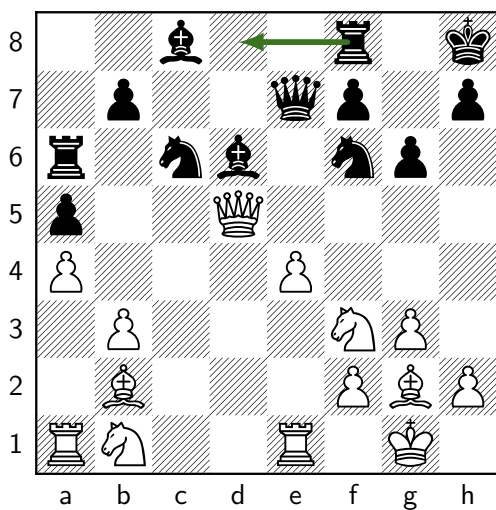


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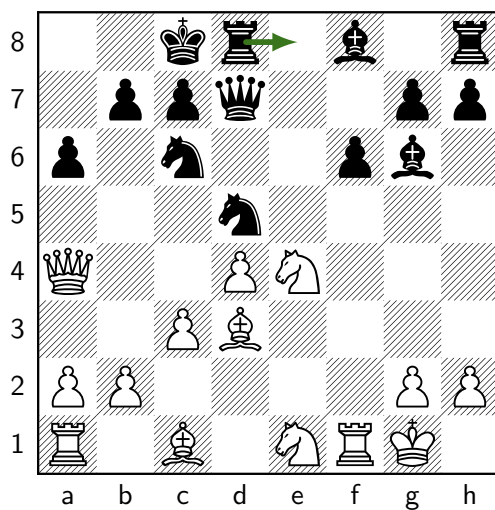


(d)

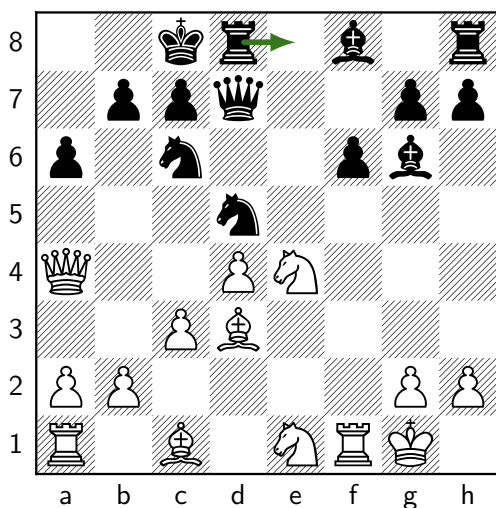
Figure 8: **Move 15, Black: Bd6.** (a) Our position; Melanie plays Bd6 (c5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#668` (ply 24) [structural distance $d=87$, QAP obj=42]: the master played Nc5; under μ our *Bc4* plays the role of that game’s *Na4* (a **cross-type** correspondence); cross-type roles: our *pa4* as their *Ra1*, our *Bc5* as their *Rc1*; transferred move c5d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2407` (ply 19) [structural distance $d=93$, QAP obj=39]: the master played Bxc5; under μ our *Bc4* plays the role of that game’s *Bd3*; cross-type roles: our *Pf7* as their *Ra8*, our *Rf8* as their *Qd8*; transferred move c5d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#250` (ply 30) [structural distance $d=85$, QAP obj=28]: the master played Qe3; under μ our *Bc4* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Ra1*, our *Ra6* as their *Pa2*; transferred move c5d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd6 (summed QAP objective 124, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



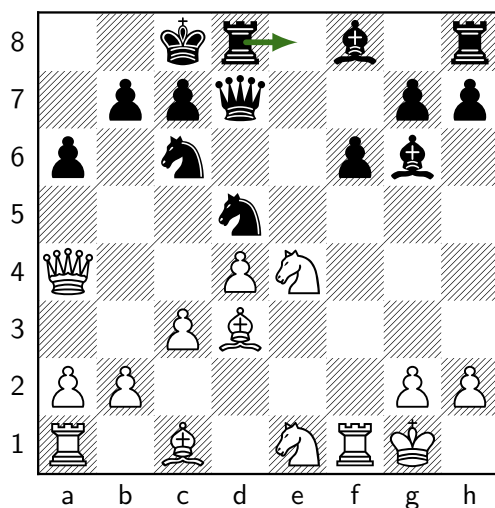
(a)



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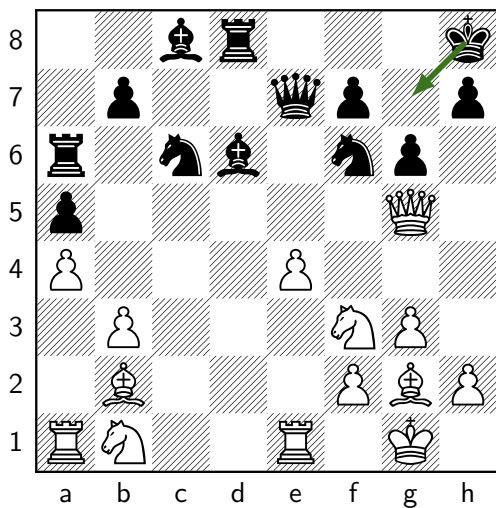


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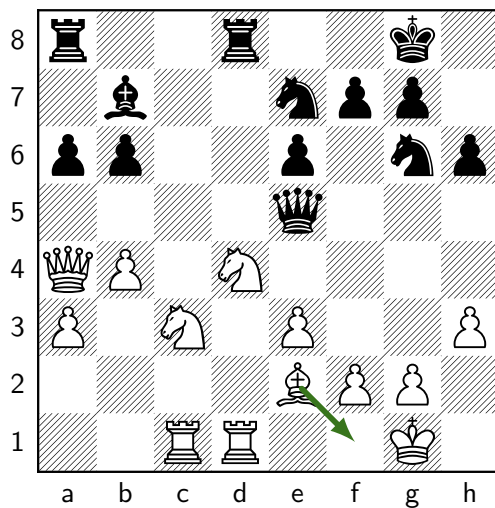


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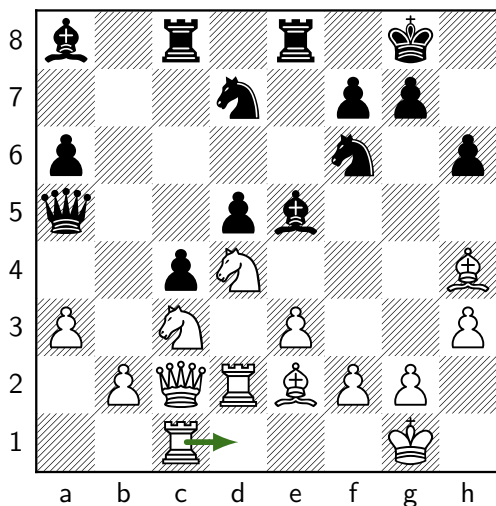
Figure 9: **Move 16, Black: Rd8.** (a) Our position; Melanie plays Rd8 (f8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#111 (ply 29) [structural distance $d=101$, QAP obj=22]: the master played Re8; under μ our Rf1 plays the role of that game’s Rd1; cross-type roles: our Ra6 as their Bf8, our Bc8 as their Rh8; transferred move f8d8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#112 (ply 29) [structural distance $d=101$, QAP obj=22]: the master played Re8; under μ our Rf1 plays the role of that game’s Rd1; cross-type roles: our Ra6 as their Bf8, our Bc8 as their Rh8; transferred move f8d8 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#2352 (ply 29) [structural distance $d=101$, QAP obj=22]: the master played Re8; under μ our Rf1 plays the role of that game’s Rd1; cross-type roles: our Ra6 as their Bf8, our Bc8 as their Rh8; transferred move f8d8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rd8 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



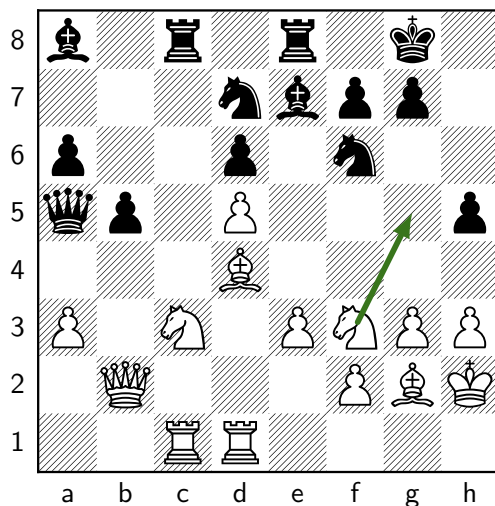
(a)



(b)

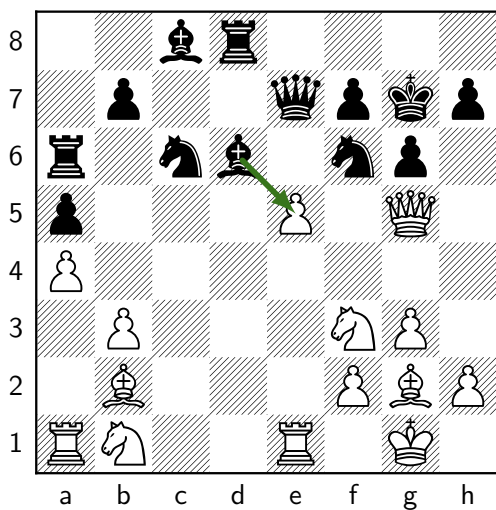


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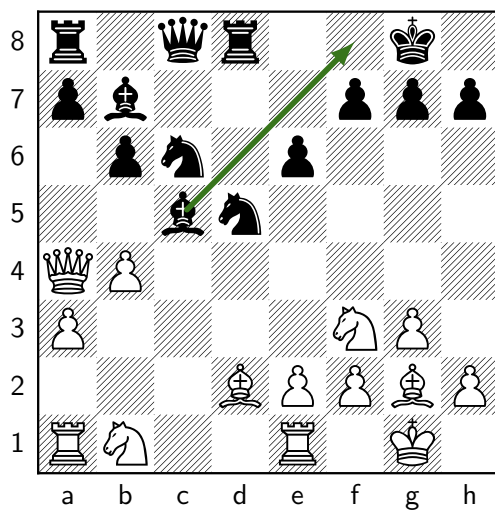


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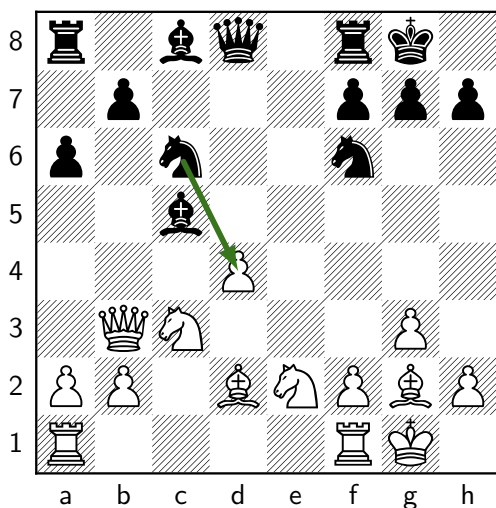
Figure 10: **Move 17, Black: Kg7**. (a) Our position; Melanie plays Kg7 (h8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3623` (ply 40) [structural distance $d=91$, QAP obj=80]: the master played Bf1; under μ our Ph2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Rc1, our Nc6 as their Pg2; transferred move h7h5. (c) Retrieved from `ccrl_top_p0000#3089` (ply 44) [structural distance $d=83$, QAP obj=73]: the master played Rcd1; under μ our Rd1 plays the role of that game’s Rc1; cross-type roles: our Nc6 as their Qc2, our re1 as their Pf2; transferred move d8d7. (d) Retrieved from `ccrl_top_p0000#1559` (ply 44) [structural distance $d=87$, QAP obj=67]: the master played Ng5; under μ our Rd1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rd1, our Pf7 as their Bg2; transferred move d8f8. Synthesis: no retrieved analog transferred to Kg7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



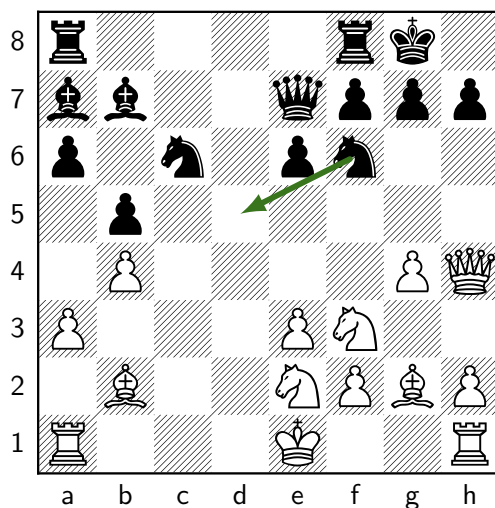
(a)



(b)

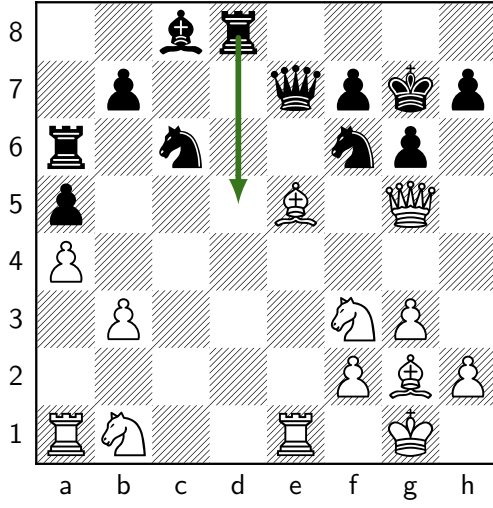


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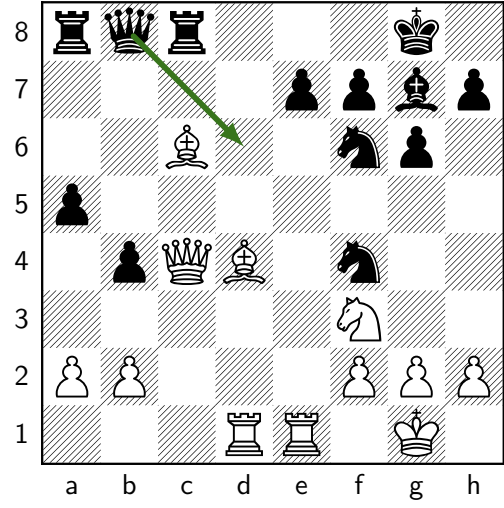


(d)

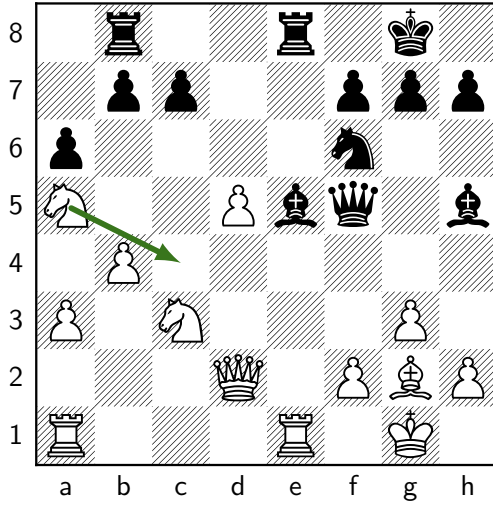
Figure 11: **Move 18, Black: Bxe5.** (a) Our position; Melanie plays Bxe5 (d6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#202` (ply 27) [structural distance $d=77$, QAP obj=80]: the master played Bf8; under μ our Bd3 plays the role of that game’s Bc4; cross-type roles: our Ph7 as their Qc8, our Nf6 as their Pa7; transferred move d6c7. (c) Retrieved from `ccr1_top_p0000#3118` (ply 27) [structural distance $d=95$, QAP obj=79]: the master played Nxd4; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our bb2 as their Ra8, our ph2 as their Bc8; transferred move c6b4. (d) Retrieved from `ccr1_top_p0001#3980` (ply 31) [structural distance $d=91$, QAP obj=45]: the master played Nd5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Bc8 as their Ra8, our Nc6 as their Ba7; transferred move f6e4. Synthesis: no retrieved analog transferred to Bxe5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



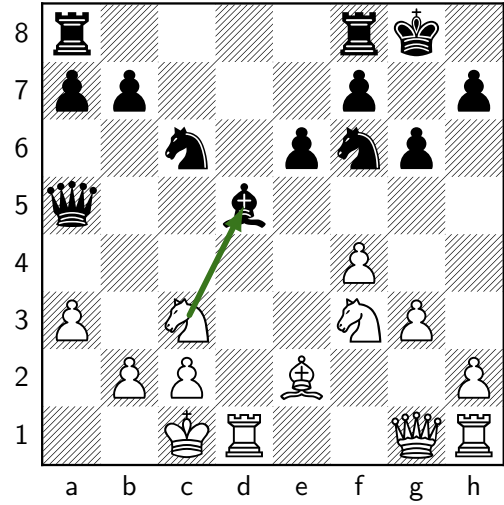
(a)



(b)

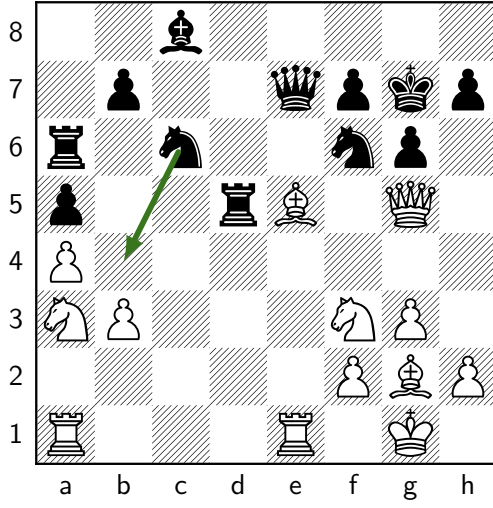


(c)

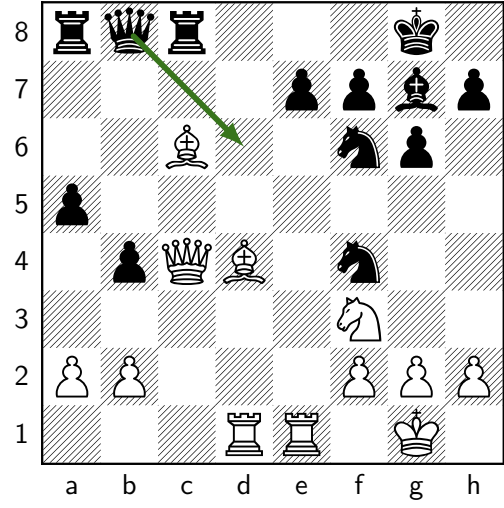


(d)

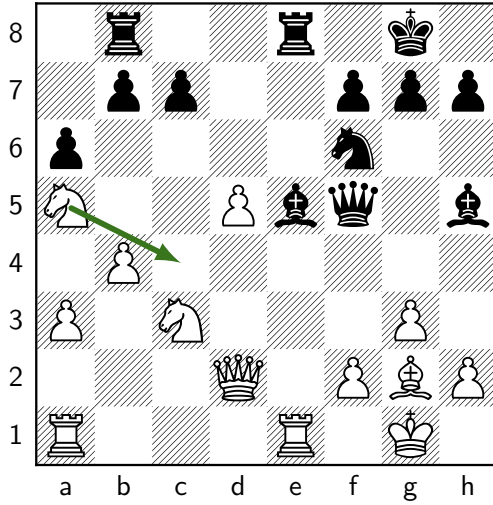
Figure 12: **Move 19, Black: Rd5.** (a) Our position; Melanie plays Rd5 (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2006` (ply 47) [structural distance $d=101$, QAP obj=83]: the master played Qd6; under μ our *Rd1* plays the role of that game’s *Qb1* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Ra8*, our *Qe7* as their *Rc8*; transferred move d8d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1463` (ply 48) [structural distance $d=96$, QAP obj=77]: the master played Nc4; under μ our *Bc1* plays the role of that game’s *Na5* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Ra1*, our *Qe7* as their *Re1*; transferred move c8f5. (d) Retrieved from `ccr1_top_p0000#3796` (ply 32) [structural distance $d=96$, QAP obj=71]: the master played Nxd5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Qe7* as their *Rd1*, our *Rd8* as their *Qg1*; transferred move c6e5. Synthesis: 1 retrieved analog transferred to Rd5 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



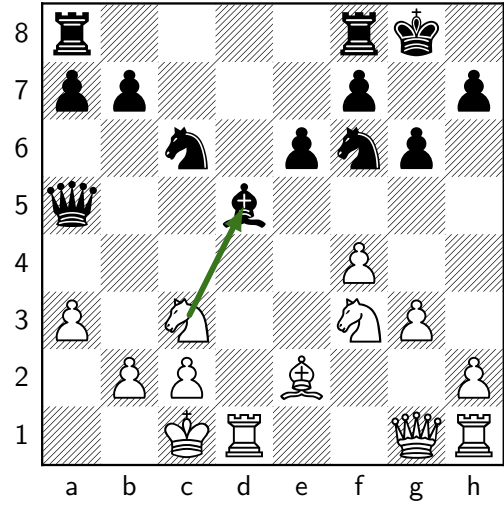
(a)



(b)

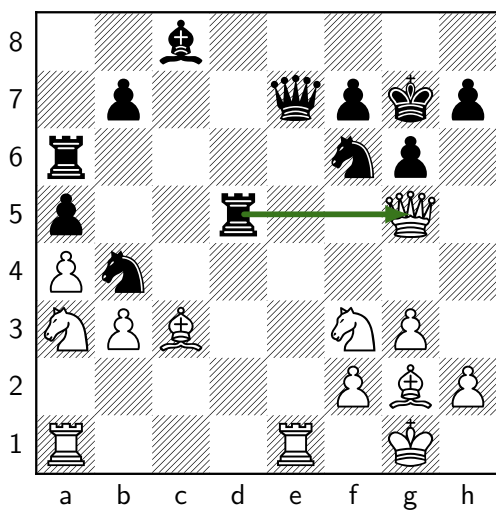


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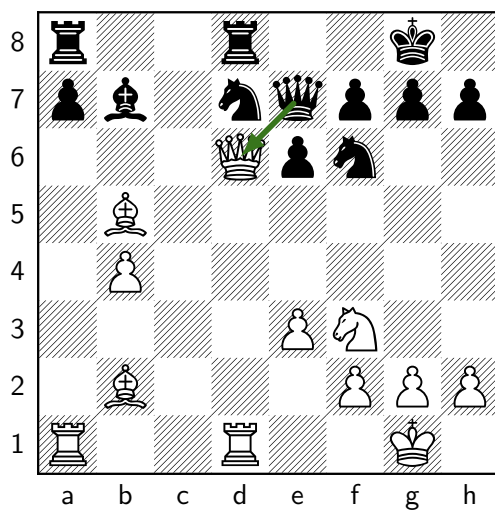


(d)

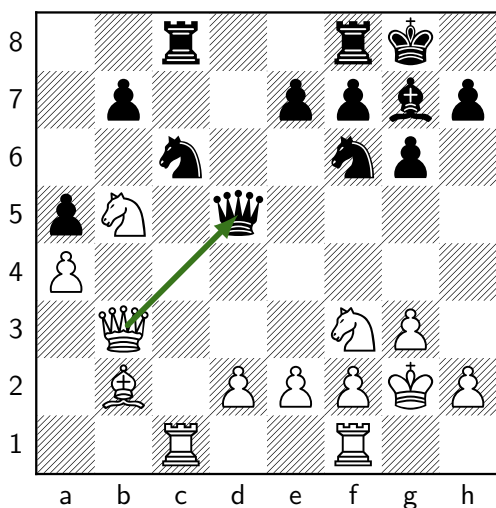
Figure 13: **Move 20, Black: Nb4.** (a) Our position; Melanie plays Nb4 (c6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2006` (ply 47) [structural distance $d=126$, QAP obj=81]: the master played Qd6; under μ our Nc3 plays the role of that game’s Qb1 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Ra8, our Qe7 as their Rc8; transferred move c6b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1463` (ply 48) [structural distance $d=132$, QAP obj=76]: the master played Nc4; under μ our Ra3 plays the role of that game’s Na5 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra1, our ph2 as their Bg2; transferred move a6a8. (d) Retrieved from `ccr1_top_p0000#3796` (ply 32) [structural distance $d=139$, QAP obj=71]: the master played Nxd5; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Qe7 as their Rd1, our Rd5 as their Qg1; transferred move c6e5. Synthesis: 1 retrieved analog transferred to Nb4 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



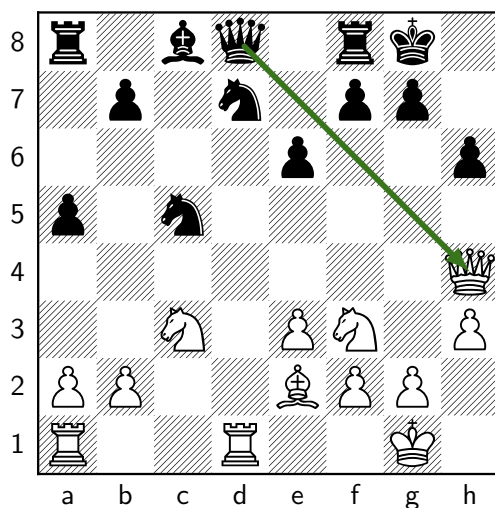
(a)



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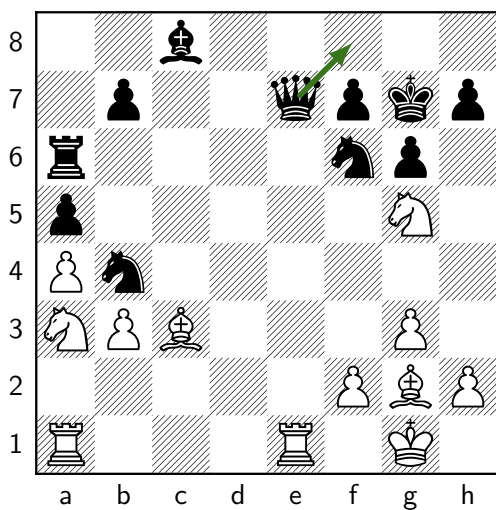


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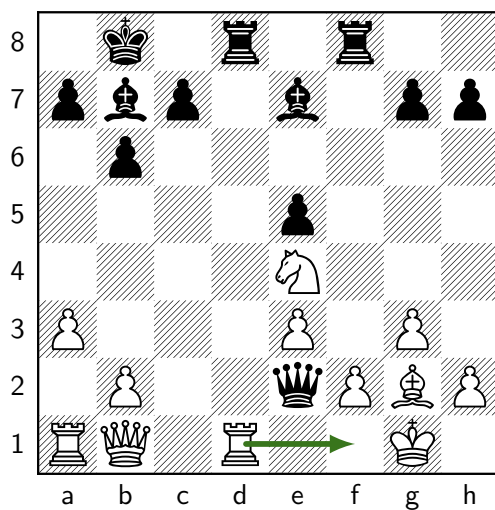


(d)

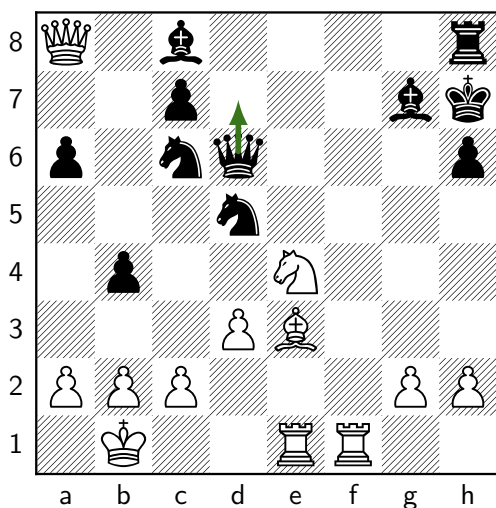
Figure 14: **Move 21, Black: Rxd5.** (a) Our position; Melanie plays Rxd5 (d5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1793` (ply 31) [structural distance $d=131$, QAP obj=103]: the master played Qxd6; under μ our $Rd4$ plays the role of that game’s $Qe2$ (a **cross-type** correspondence); cross-type roles: our $ph2$ as their $Ra8$, our $Pf7$ as their $Rd8$; transferred move d5g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2260` (ply 30) [structural distance $d=109$, QAP obj=78]: the master played Qxd5; under μ our $Rd4$ plays the role of that game’s $Qb3$ (a **cross-type** correspondence); cross-type roles: our $Pa5$ as their $Rc1$, our $Pf7$ as their $Rf1$; transferred move d5g5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2286` (ply 31) [structural distance $d=134$, QAP obj=29]: the master played Qxh4; under μ our $Rd4$ plays the role of that game’s $Qd1$ (a **cross-type** correspondence); cross-type roles: our $Pa5$ as their $Bc8$, our $Nf6$ as their $Rf8$; transferred move d5g5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rxd5 (summed QAP objective 210, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



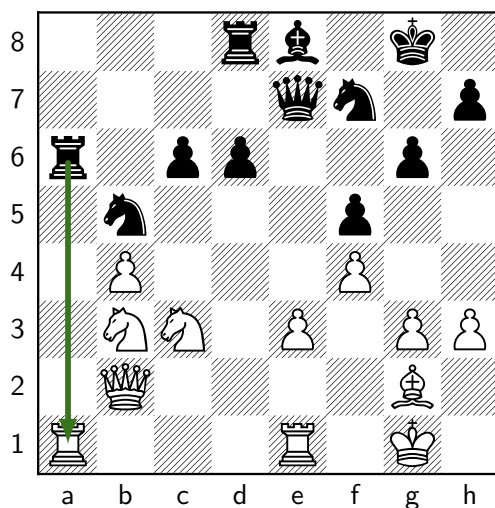
(a)



(b)

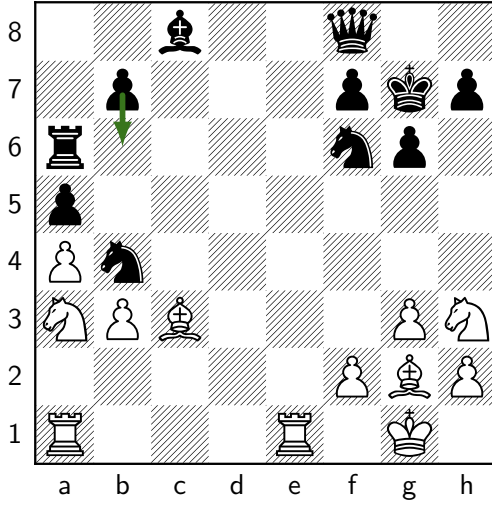


(c)

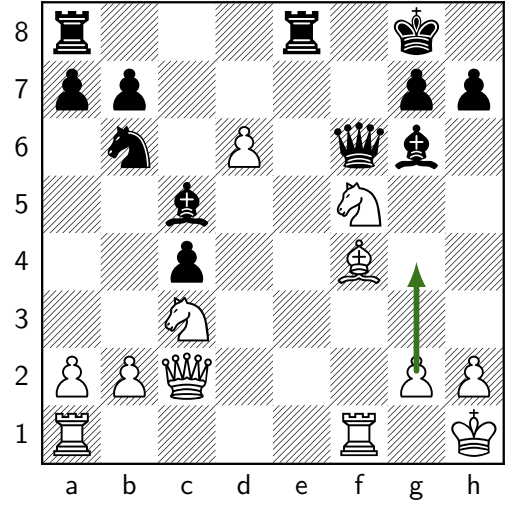


(d)

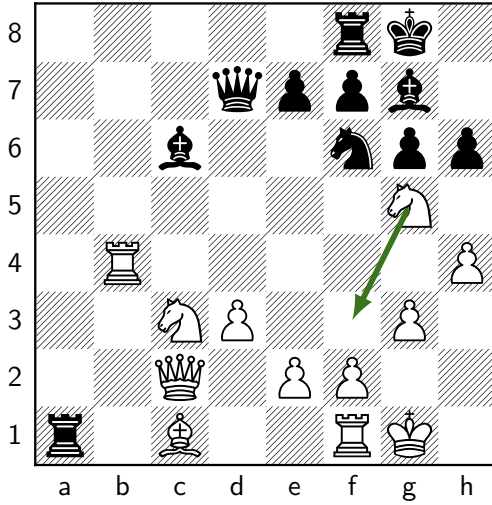
Figure 15: **Move 22, Black: Qf8.** (a) Our position; Melanie plays Qf8 (e7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#146` (ply 42) [structural distance $d=128$, QAP obj=58]: the master played Rf1; under μ our Nb5 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our na3 as their Ra1, our bc3 as their qe2; transferred move b4c6. (c) Retrieved from `ccr1_top_p0002#2149` (ply 53) [structural distance $d=124$, QAP obj=40]: the master played Qd7; under μ our Qe2 plays the role of that game’s Qd3; cross-type roles: our bc3 as their qa8, our Nf6 as their Bc8; transferred move e7e8. (d) Retrieved from `ccr1_top_p0000#645` (ply 59) [structural distance $d=129$, QAP obj=28]: the master played Rxa1; under μ our Qe2 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Be8, our bc3 as their Qe7; transferred move e7e1. Synthesis: no retrieved analog transferred to Qf8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



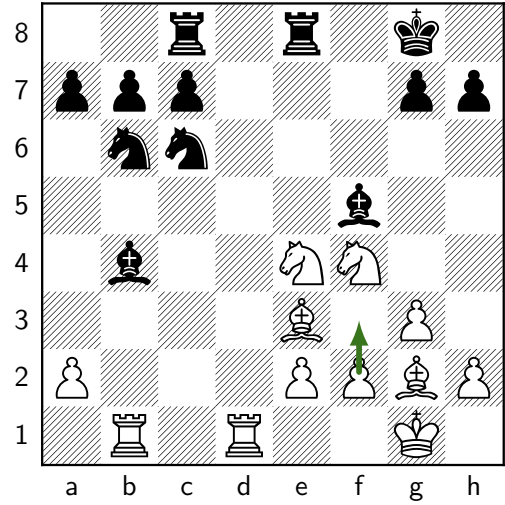
(a)



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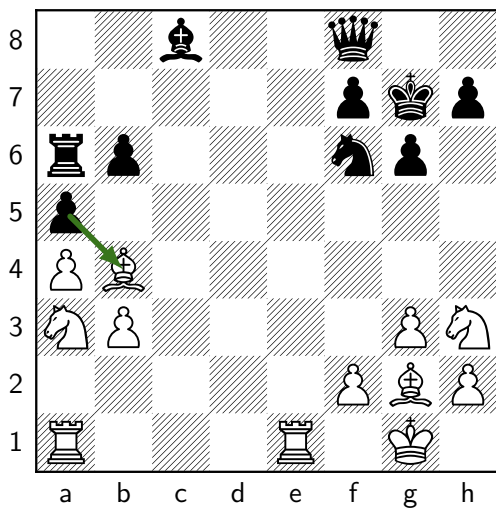


(c)

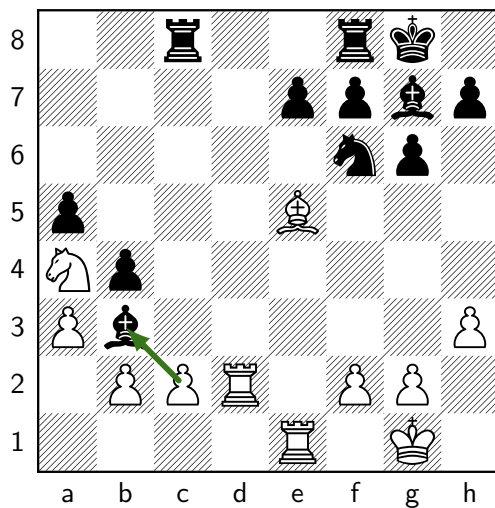


(d)

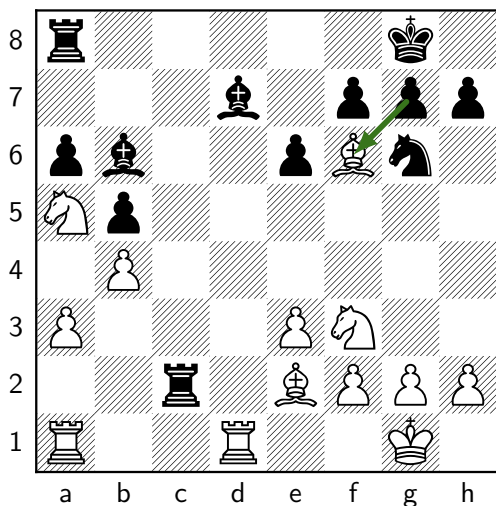
Figure 16: **Move 23, Black: b6**. (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2714` (ply 40) [structural distance $d=116$, QAP obj=47]: the master played g4; under μ our *Pb2* plays the role of that game’s *Pg2*; cross-type roles: our *Bc8* as their *Ra1*, our *Qf8* as their *Rf1*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2575` (ply 44) [structural distance $d=116$, QAP obj=44]: the master played Nf3; under μ our *Nb5* plays the role of that game’s *Ng5*; cross-type roles: our *bc3* as their *ra1*, our *Nf6* as their *Bc1*; transferred move b4c6. (d) Retrieved from `ccrl_top_p0002#3199` (ply 38) [structural distance $d=116$, QAP obj=28]: the master played f3; under μ our *Ph2* plays the role of that game’s *Pf2*; cross-type roles: our *Bc8* as their *Rb1*, our *pa4* as their *Rd1*; transferred move h7h5. Synthesis: 1 retrieved analog transferred to b6 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



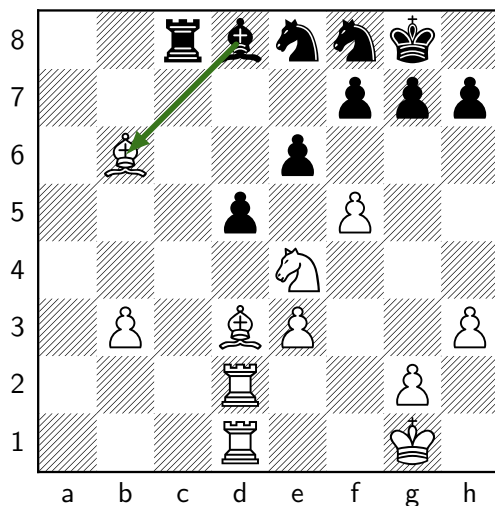
(a)



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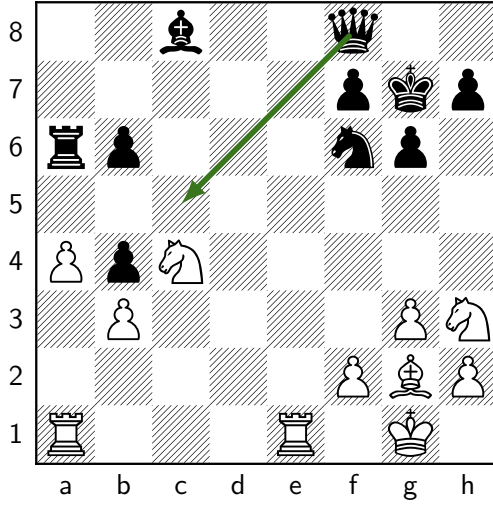


(c)

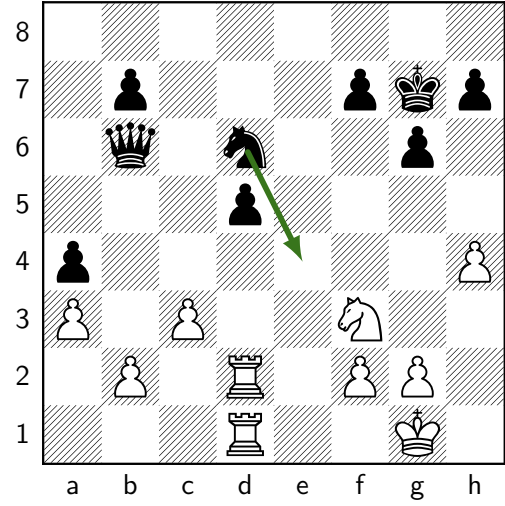


(d)

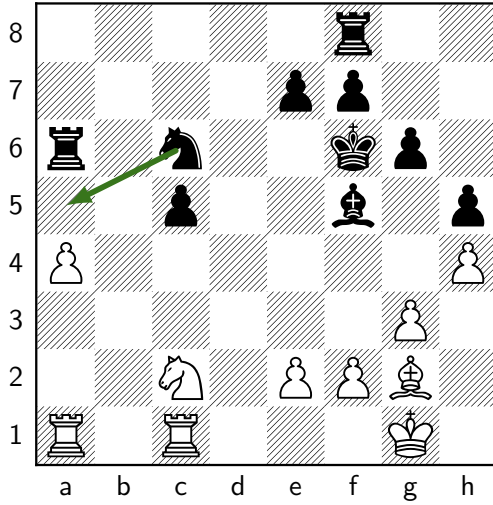
Figure 17: **Move 24, Black: axb4.** (a) Our position; Melanie plays axb4 (a5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3641` (ply 40) [structural distance $d=101$, QAP obj=55]: the master played cxb3; under μ our $Pa4$ plays the role of that game’s $Pc2$; cross-type roles: our $Pb6$ as their $Re1$, our $Bc8$ as their $Pb2$; transferred move a5b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1326` (ply 35) [structural distance $d=106$, QAP obj=38]: the master played gxf6; under μ our $Pa4$ plays the role of that game’s $Pg2$; cross-type roles: our $na3$ as their $Ra8$, our $pg3$ as their $Bd7$; transferred move a5b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3652` (ply 59) [structural distance $d=101$, QAP obj=32]: the master played Bxb6; under μ our $Pa4$ plays the role of that game’s $Bd1$ (a **cross-type** correspondence); cross-type roles: our $re1$ as their $Ne8$, our $ph2$ as their $Nf8$; transferred move a5b4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to axb4 (summed QAP objective 146, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



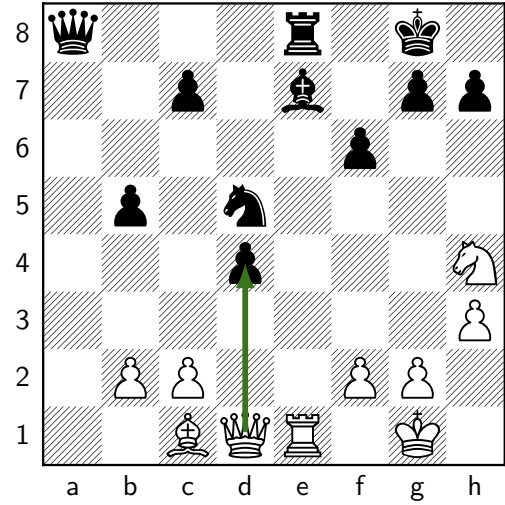
(a)



(b)

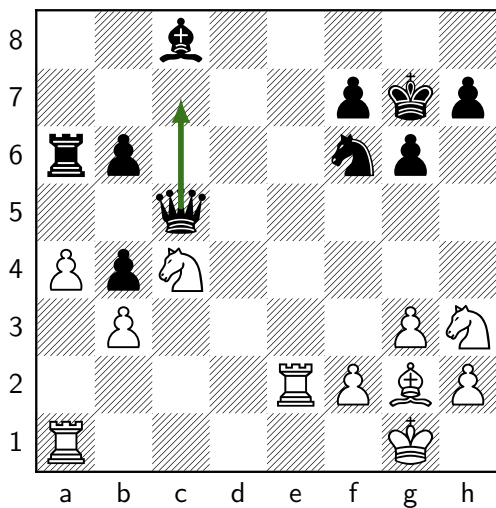


(c)

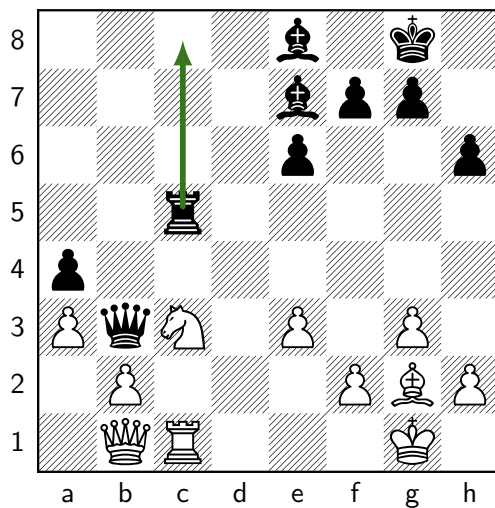


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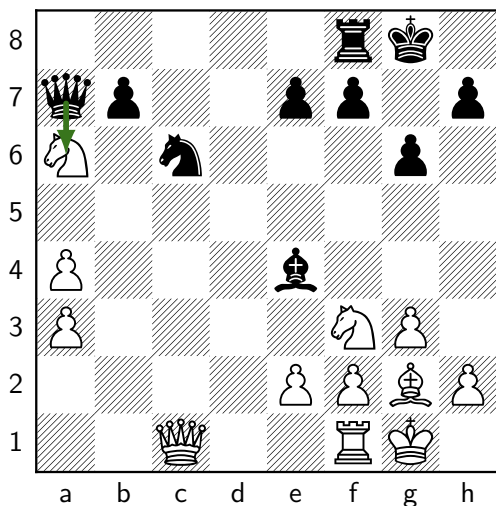
Figure 18: **Move 25, Black: Qc5.** (a) Our position; Melanie plays Qc5 (f8c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2980` (ply 63) [structural distance $d=92$, QAP obj=43]: the master played Ne4; under μ our Qf1 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our nc4 as their Qb6, our Nf6 as their Pg6; transferred move f8c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1563` (ply 59) [structural distance $d=95$, QAP obj=59]: the master played Na5; under μ our Pb3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Pb4 as their Rf8, our Qf8 as their Pe7; transferred move b6b5. (d) Retrieved from `ccr1_top_p0000#3692` (ply 38) [structural distance $d=94$, QAP obj=59]: the master played Qxd4; under μ our Ra3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Bc1, our Nf6 as their Pg2; transferred move a6a7. Synthesis: 1 retrieved analog transferred to Qc5 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



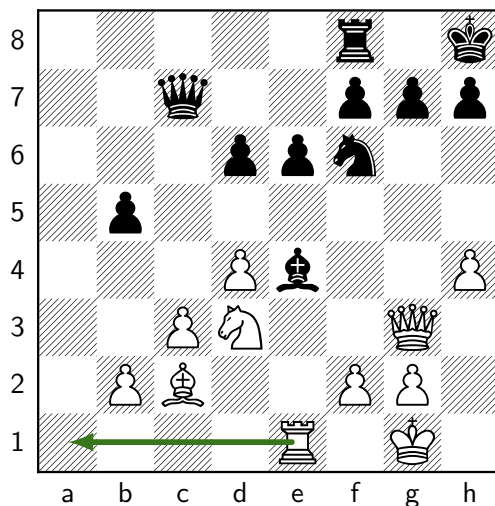
(a)



(b)



(c)



(d)

Figure 19: **Move 26, Black: Qc7**. (a) Our position; Melanie plays Qc7 (c5c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2982` (ply 51) [structural distance $d=86$, QAP obj=76]: the master played Rc8; under μ our Qc4 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Ra6 as their Be8, our pb3 as their Be7; transferred move c5c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3881` (ply 37) [structural distance $d=96$, QAP obj=65]: the master played Qxa6; under μ our Qc4 plays the role of that game’s Qa2; cross-type roles: our Pf7 as their Rf8, our Qc5 as their Pb7; transferred move c5c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2475` (ply 52) [structural distance $d=99$, QAP obj=47]: the master played Ra1; under μ our Qc4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pb2, our Pb4 as their Bc2; transferred move c5c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc7 (summed QAP objective 231, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

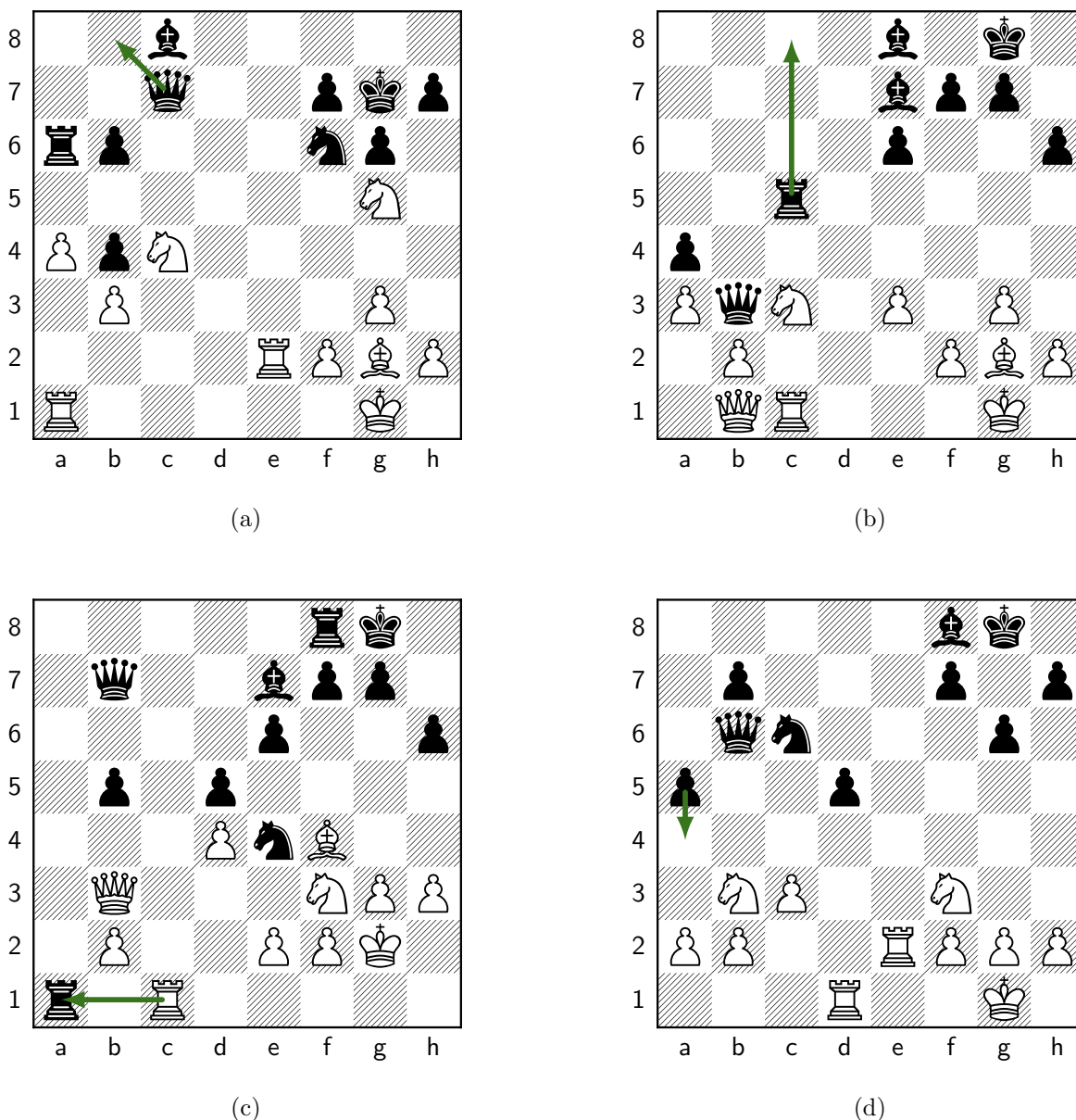
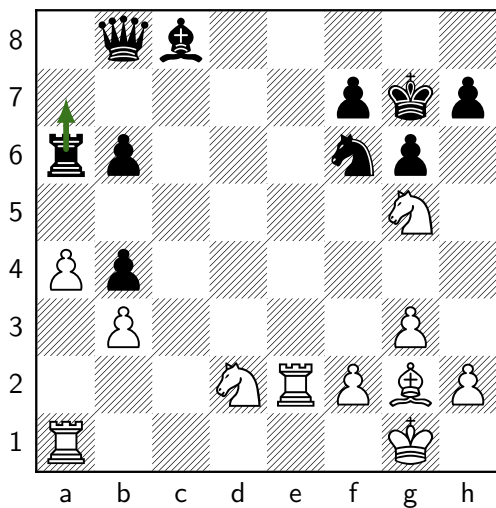
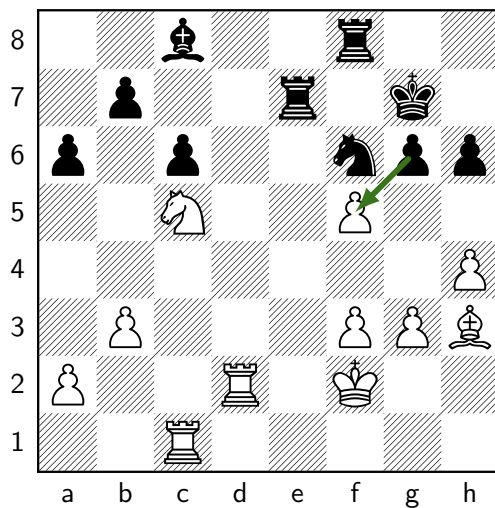


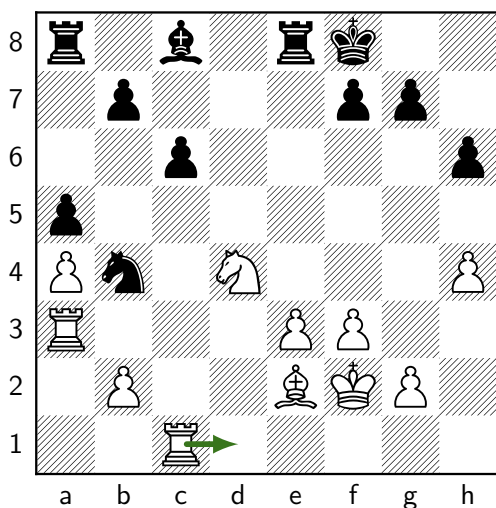
Figure 20: **Move 27, Black: Qb8.** (a) Our position; Melanie plays Qb8 (c7b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2982` (ply 51) [structural distance $d=92$, QAP obj=43]: the master played Rc8; under μ our Qc2 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Ra6 as their Be8, our Ph7 as their Be7; transferred move c7b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1112` (ply 48) [structural distance $d=83$, QAP obj=32]: the master played Rxa1; under μ our Qc2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our nc4 as their ra1, our Ra6 as their Qb3; transferred move c7b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2980` (ply 39) [structural distance $d=68$, QAP obj=63]: the master played a4; under μ our Pb3 plays the role of that game’s Pa4; cross-type roles: our Ra6 as their Bf8, our bg2 as their Pb7; transferred move b6b5. Synthesis: 2 retrieved analogs transferred to Qb8 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



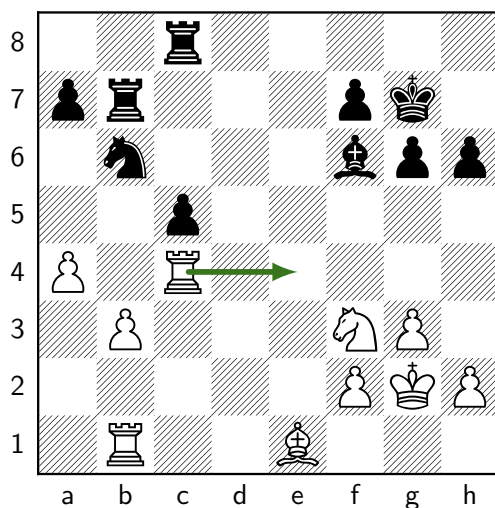
(a)



(b)



(c)



(d)

Figure 21: **Move 28, Black: Ra7.** (a) Our position; Melanie plays Ra7 (a6a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1` (ply 57) [structural distance $d=72$, QAP obj=58]: the master played **gxf5**; under μ our *Ra3* plays the role of that game’s *Pg3* (a **cross-type** correspondence); cross-type roles: our *Ra6* as their *Bc8*, our *Bc8* as their *Rf8*; transferred move a6a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#876` (ply 50) [structural distance $d=60$, QAP obj=46]: the master played **Rd1**; under μ our *Ra3* plays the role of that game’s *Rc1*; cross-type roles: our *nd2* as their *Be2*, our *pg3* as their *Ra3*; transferred move a6a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2163` (ply 56) [structural distance $d=75$, QAP obj=16]: the master played **Re4**; under μ our *Ra3* plays the role of that game’s *Rc4*; cross-type roles: our *Bc8* as their *Rb1*, our *pg3* as their *Be1*; transferred move a6a7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ra7 (summed QAP objective 124, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

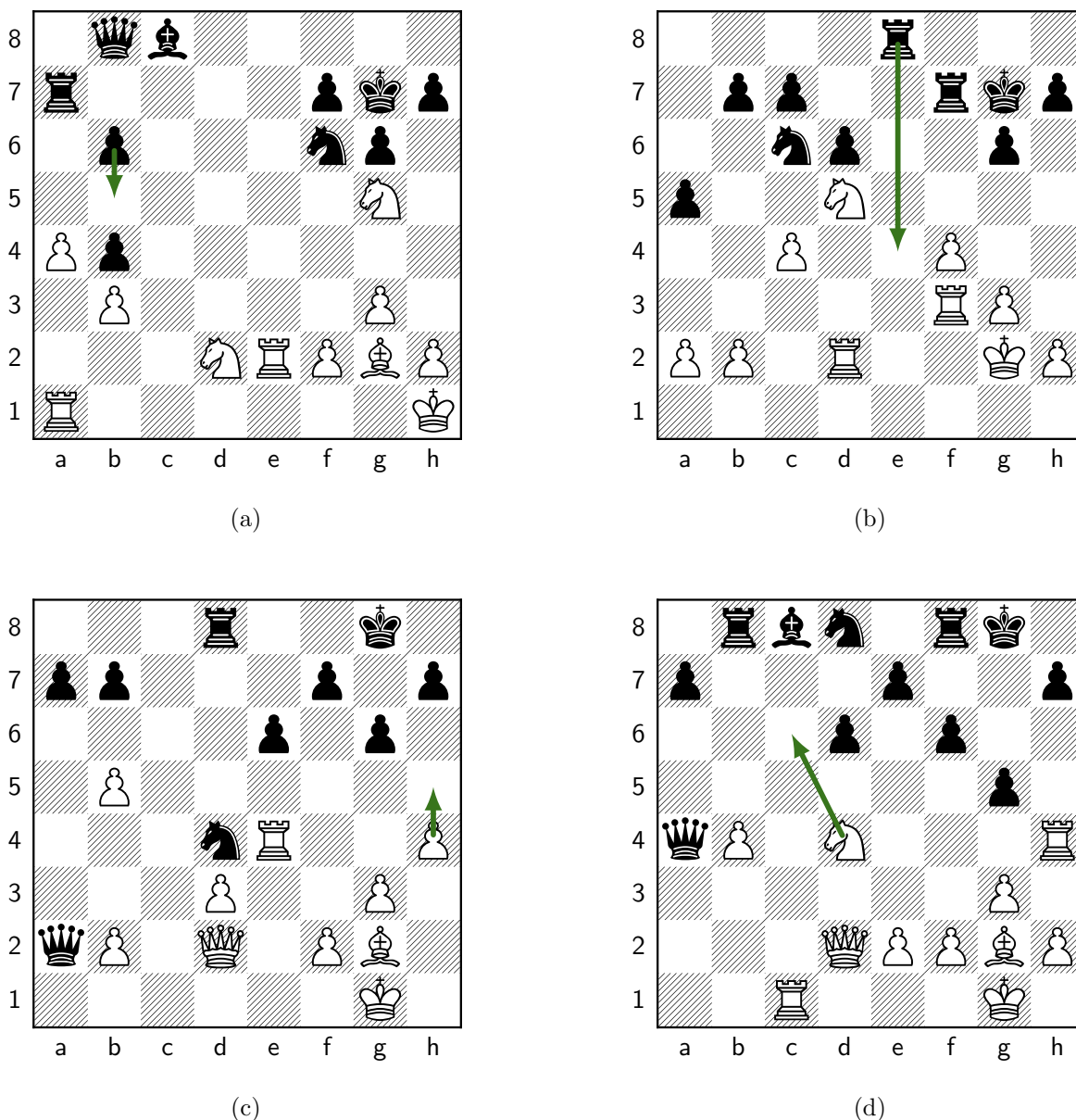


Figure 22: **Move 29, Black: b5**. (a) Our position; Melanie plays b5 (b6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2971` (ply 51) [structural distance $d=81$, QAP obj=61]: the master played Re4 ; under μ our Pb3 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our re2 as their Pd6 , our Bc8 as their Pg6 ; transferred move b6b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3241` (ply 60) [structural distance $d=81$, QAP obj=14]: the master played h5 ; under μ our Pb3 plays the role of that game’s Ph4 ; cross-type roles: our Ra7 as their qa2 , our ra1 as their Qd2 ; transferred move b6b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2424` (ply 54) [structural distance $d=79$, QAP obj=10]: the master played Nc6 ; under μ our Pb3 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Qd2 , our Qb8 as their Pe2 ; transferred move b6b5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to b5 (summed QAP objective 145, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

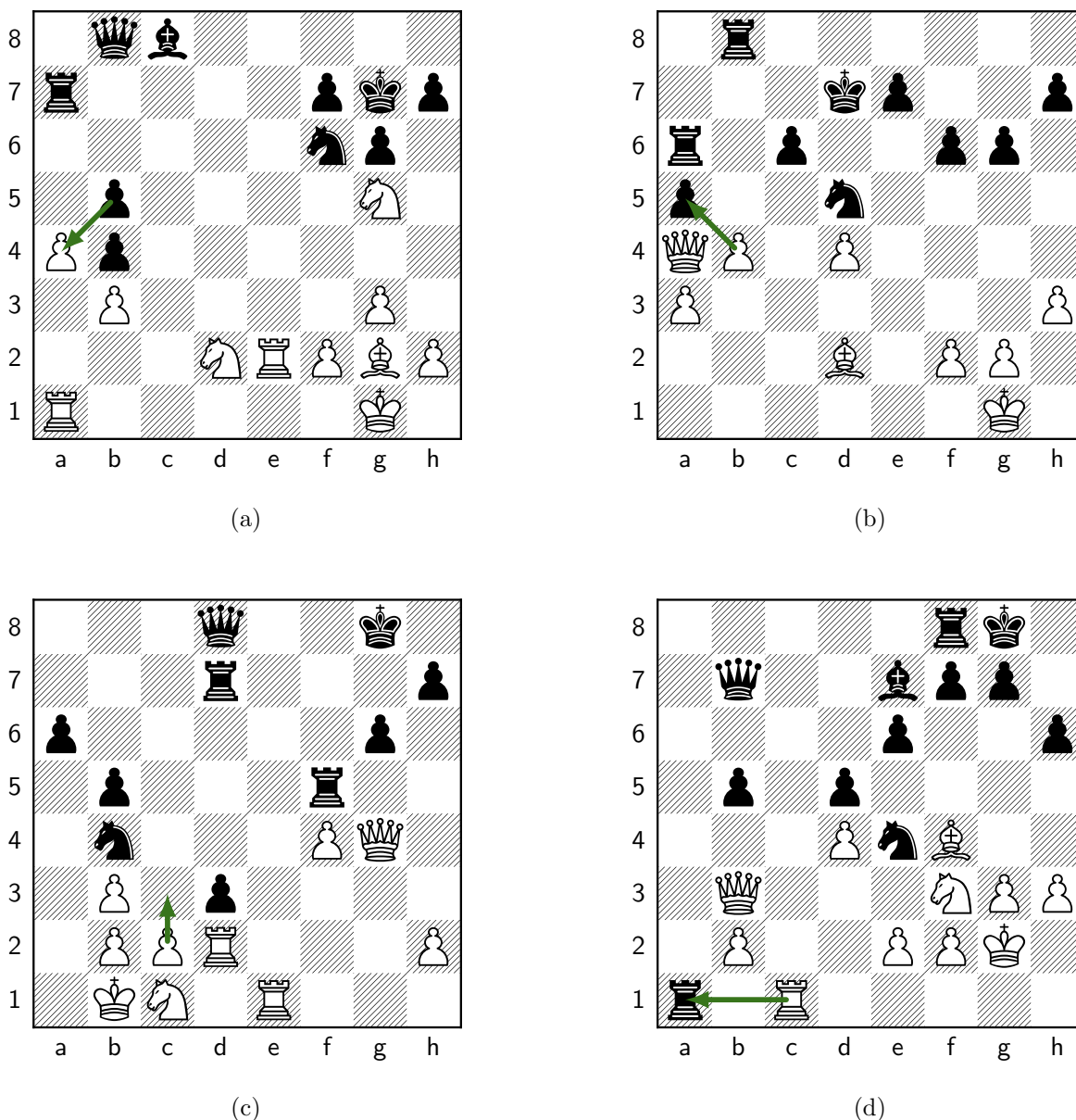
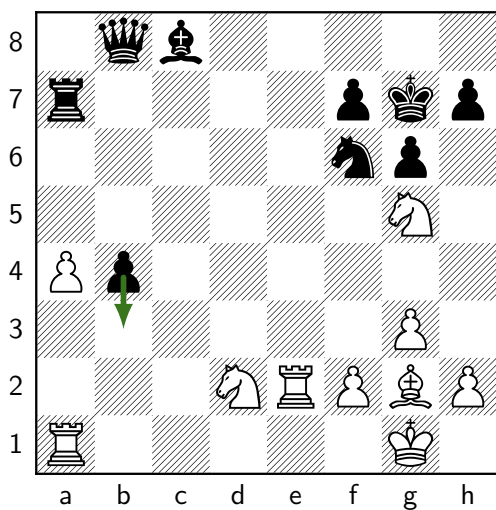
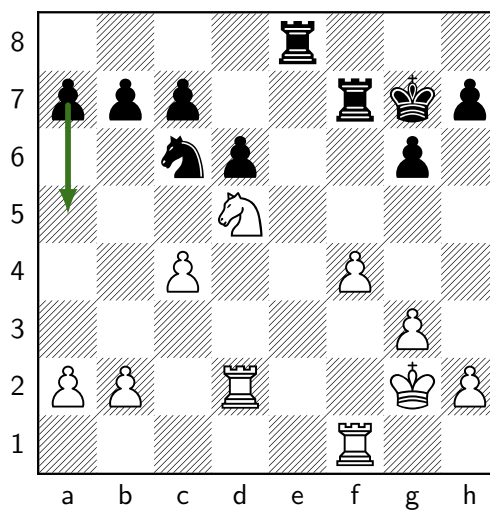


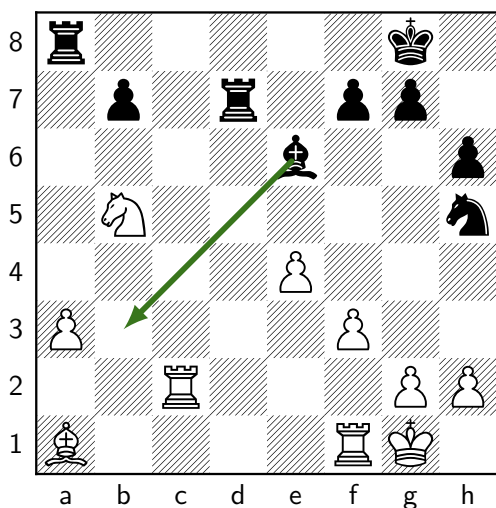
Figure 23: **Move 30, Black: bxa4.** (a) Our position; Melanie plays **bxa4** (**b5a4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3952` (ply 54) [structural distance $d=76$, QAP obj=88]: the master played **bxa5**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our Qb_8 as their Pa_3 , our Ra_7 as their Qa_4 ; transferred move **b5a4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2451` (ply 52) [structural distance $d=86$, QAP obj=41]: the master played **c3**; under μ our Pb_4 plays the role of that game’s Pc_2 ; cross-type roles: our Ra_7 as their Nc_1 , our Qb_8 as their Re_1 ; transferred move **b5a4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1112` (ply 48) [structural distance $d=91$, QAP obj=28]: the master played **Rxa1**; under μ our Pb_4 plays the role of that game’s Rc_1 (a **cross-type** correspondence); cross-type roles: our pa_4 as their ra_1 , our Nf_6 as their Pf_2 ; transferred move **b5a4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **bxa4** (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



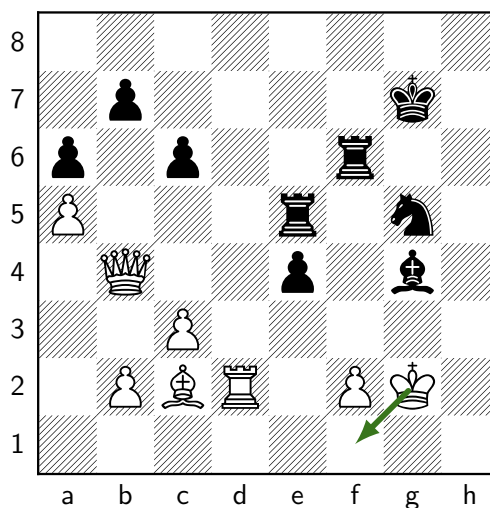
(a)



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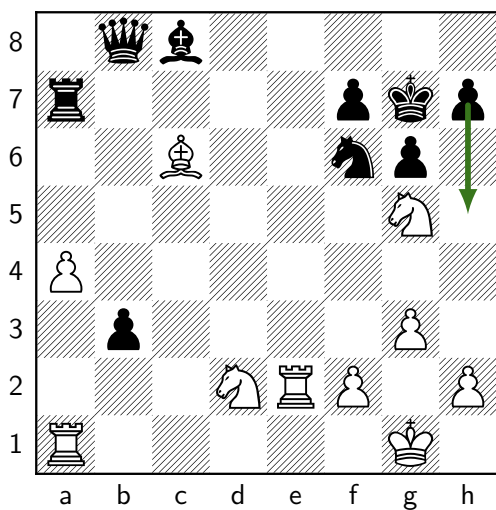


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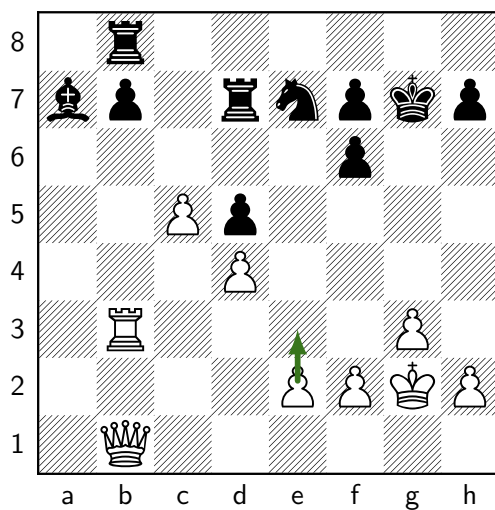


(d)

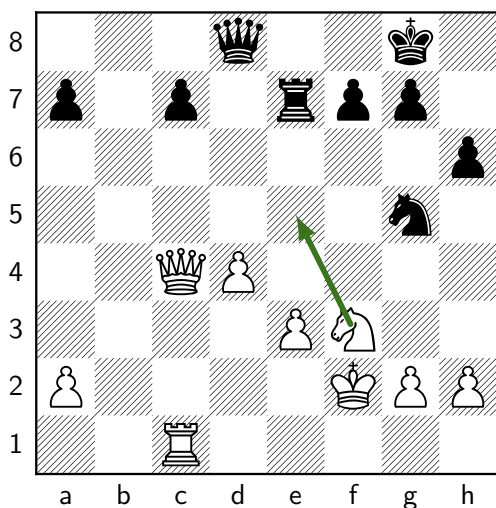
Figure 24: **Move 31, Black: b3**. (a) Our position; Melanie plays b3 (b4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2971` (ply 49) [structural distance $d=82$, QAP obj=56]: the master played a5; under μ our $Pb5$ plays the role of that game’s $Pa2$; cross-type roles: our $Qb8$ as their $Re8$, our $Bc8$ as their $Pb7$; transferred move b4b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1834` (ply 53) [structural distance $d=83$, QAP obj=32]: the master played Bb3; under μ our $Pb5$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $Qb8$ as their $Pb7$, our $Ph7$ as their $Rd7$; transferred move b4b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1206` (ply 74) [structural distance $d=83$, QAP obj=68]: the master played Kf1; under μ our $Kg2$ plays the role of that game’s $Kg2$; cross-type roles: our $Ra7$ as their $Bc2$, our $Nf6$ as their $Rd2$; transferred move g7f8. Synthesis: 2 retrieved analogs transferred to b3 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



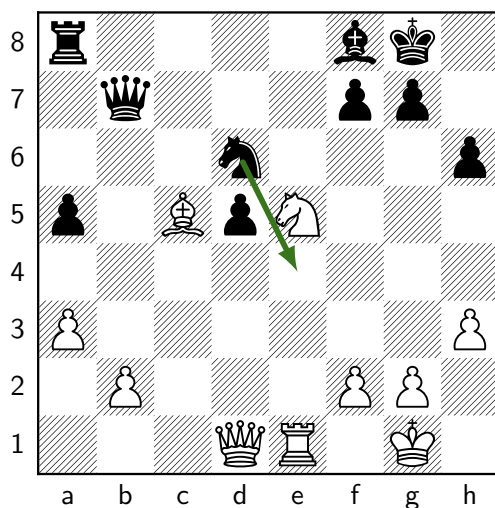
(a)



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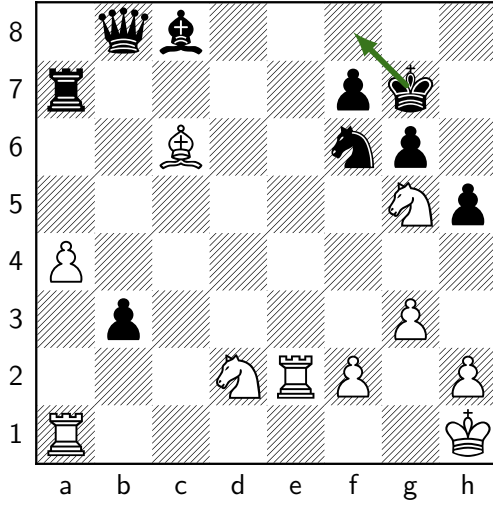


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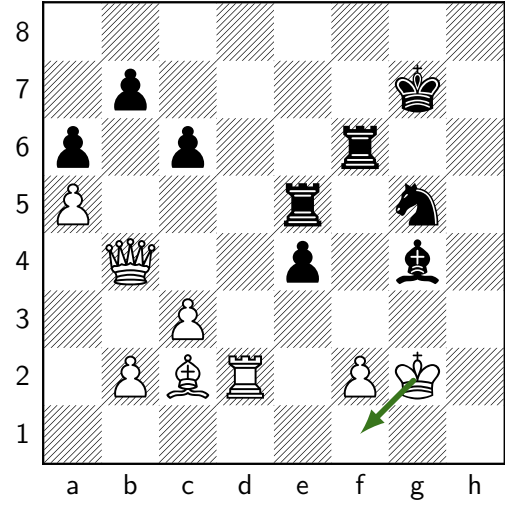


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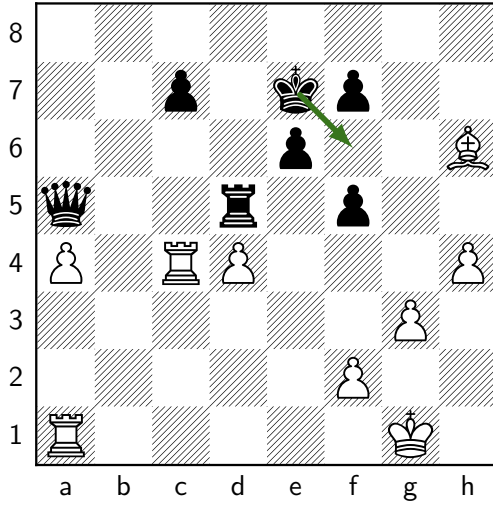
Figure 25: **Move 32, Black: h5**. (a) Our position; Melanie plays h5 (h7h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1709` (ply 74) [structural distance $d=87$, QAP obj=60]: the master played e3; under μ our *Ph2* plays the role of that game’s *Pe2*; cross-type roles: our *Nf6* as their *Ph2*, our *re2* as their *Pg3*; transferred move h7h5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2661` (ply 50) [structural distance $d=90$, QAP obj=36]: the master played Ne5; under μ our *Ph2* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Rc1*, our *Qb8* as their *Pa2*; transferred move h7h5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2320` (ply 59) [structural distance $d=76$, QAP obj=21]: the master played Ne4; under μ our *Ph2* plays the role of that game’s *Nd3* (a **cross-type** correspondence); cross-type roles: our *Qb8* as their *Ra8*, our *Pf7* as their *Bf8*; transferred move h7h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to h5 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



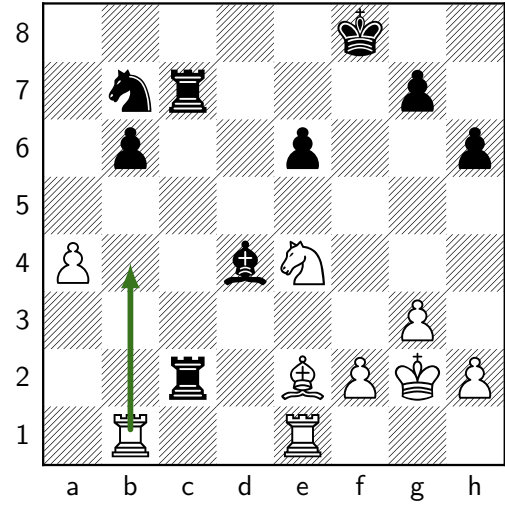
(a)



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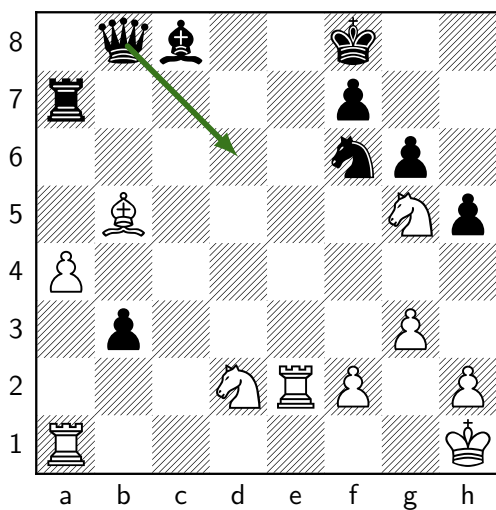


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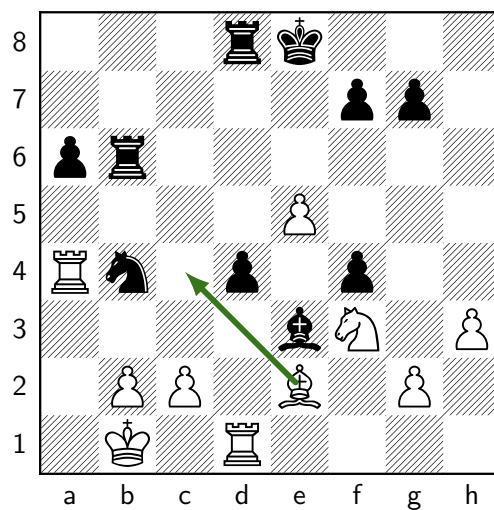


(d)

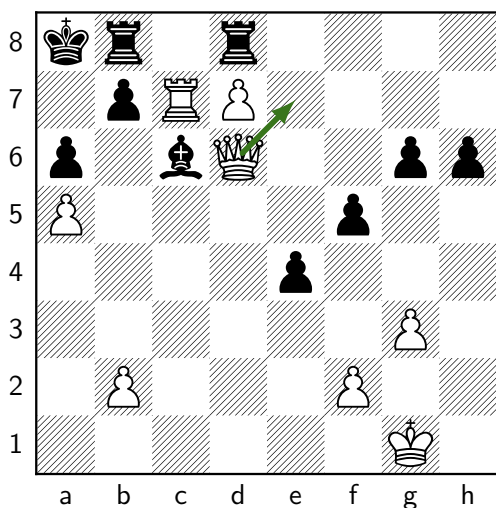
Figure 26: **Move 33, Black: Kf8**. (a) Our position; Melanie plays Kf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1206` (ply 74) [structural distance $d=78$, QAP obj=71]: the master played Kf1; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Qb8 as their Pb2, our Ra7 as their Bc2; transferred move g7f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2283` (ply 75) [structural distance $d=76$, QAP obj=33]: the master played Kf6; under μ our Kg2 plays the role of that game’s Ke2; cross-type roles: our Nf6 as their Pf7, our re2 as their Pe6; transferred move g7h6. (d) Retrieved from `ccrl_top_p0000#228` (ply 60) [structural distance $d=83$, QAP obj=32]: the master played Rb4; under μ our Ra2 plays the role of that game’s Rb1; cross-type roles: our Qb8 as their Re1, our nd2 as their rc2; transferred move a7a5. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



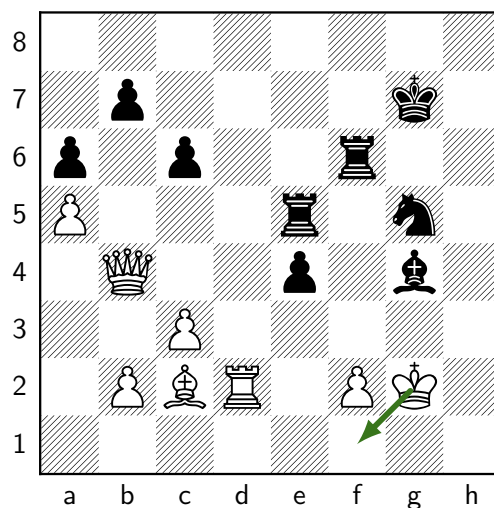
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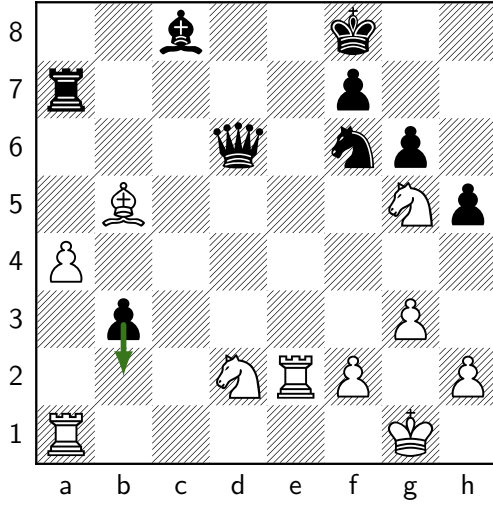


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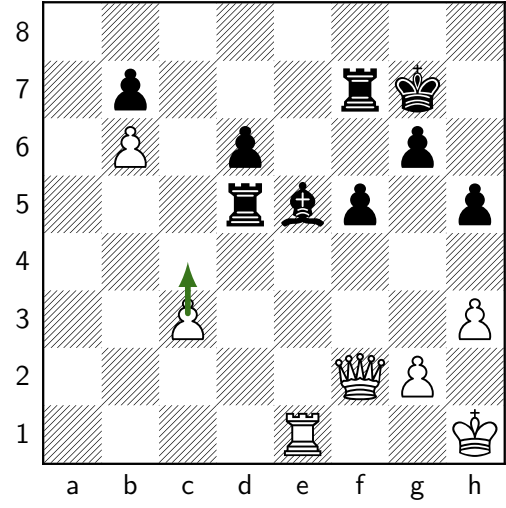


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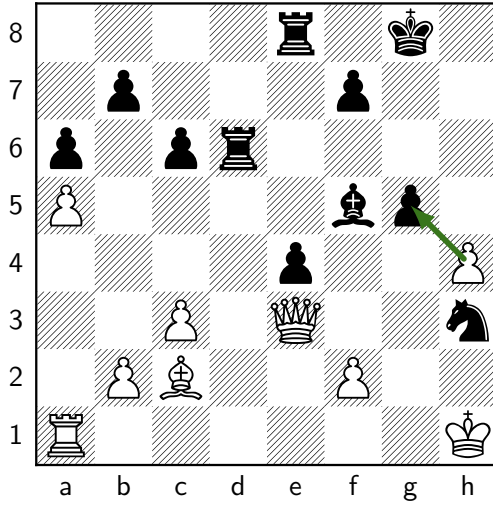
Figure 27: **Move 34, Black: Qd6**. (a) Our position; Melanie plays Qd6 (b8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#695` (ply 58) [structural distance $d=73$, QAP obj=29]: the master played Bc4; under μ our Qb1 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pb2, our Nf6 as their Pc2; transferred move b8d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#611` (ply 82) [structural distance $d=71$, QAP obj=18]: the master played Qe7; under μ our Qb1 plays the role of that game’s Qd6; cross-type roles: our Bc8 as their Pb2, our ng5 as their pf5; transferred move b8d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1206` (ply 74) [structural distance $d=63$, QAP obj=43]: the master played Kf1; under μ our Kf1 plays the role of that game’s Kg2; cross-type roles: our Qb8 as their Pb2, our Ra7 as their Bc2; transferred move f8g7. Synthesis: 2 retrieved analogs transferred to Qd6 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



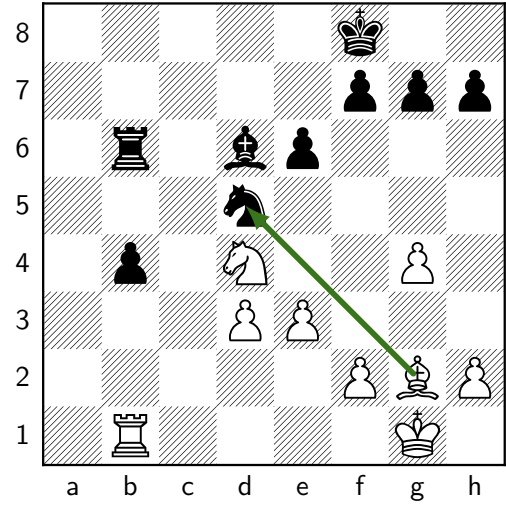
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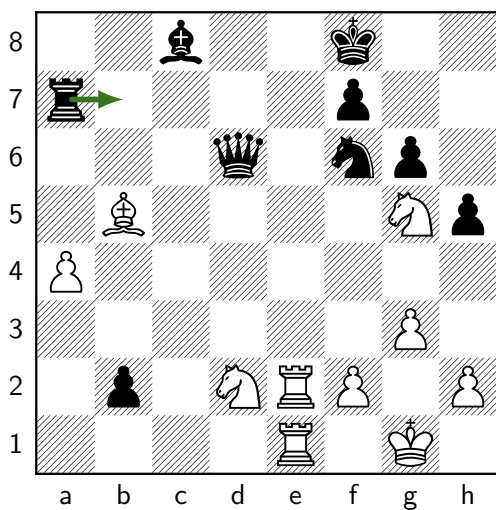


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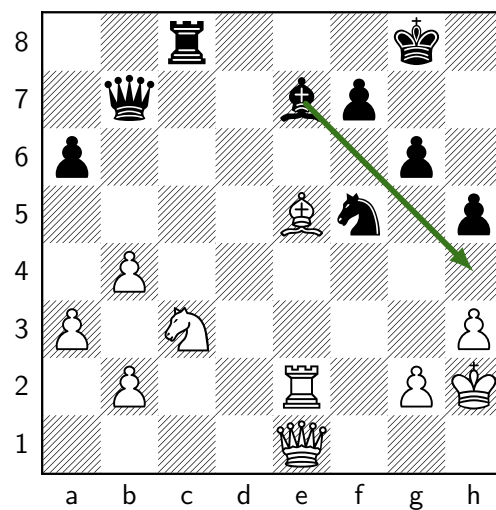


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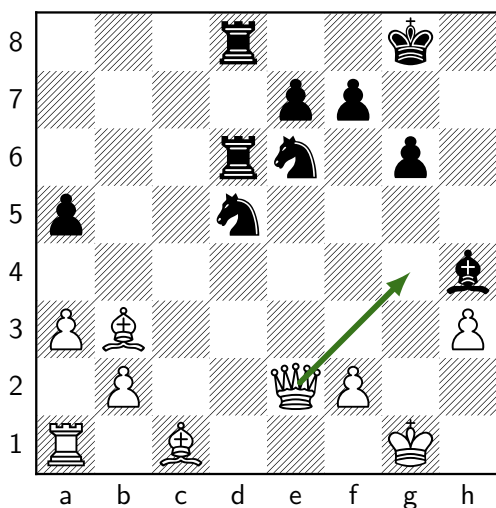
Figure 28: **Move 35, Black: b2**. (a) Our position; Melanie plays b2 (b3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2902` (ply 68) [structural distance $d=73$, QAP obj=79]: the master played c4; under μ our *Pb6* plays the role of that game’s *Pc3*; cross-type roles: our *Qd6* as their *Re1*, our *Ra7* as their *Qf2*; transferred move b3b2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1206` (ply 60) [structural distance $d=66$, QAP obj=60]: the master played hxg5; under μ our *Pb6* plays the role of that game’s *Ph4*; cross-type roles: our *Ph5* as their *Ra1*, our *Qd6* as their *Bc2*; transferred move b3b2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3050` (ply 56) [structural distance $d=76$, QAP obj=75]: the master played Bxd5; under μ our *Qd3* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Pd3*, our *re2* as their *Pg4*; transferred move d6f4. Synthesis: 2 retrieved analogs transferred to b2 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



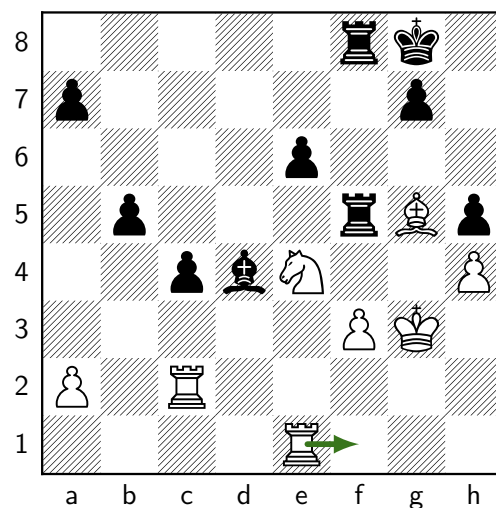
(a)



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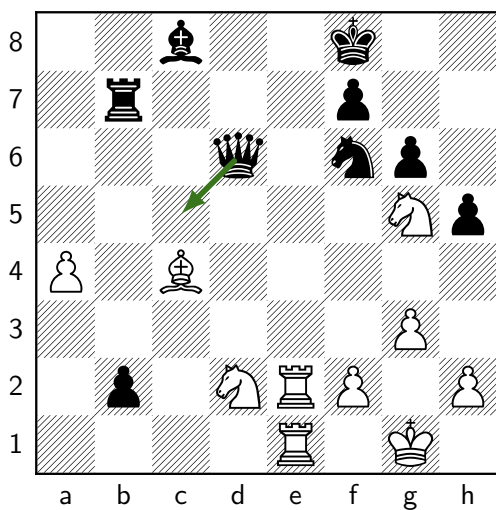


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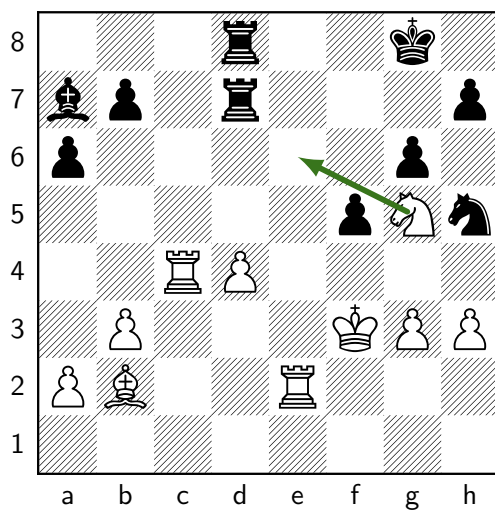


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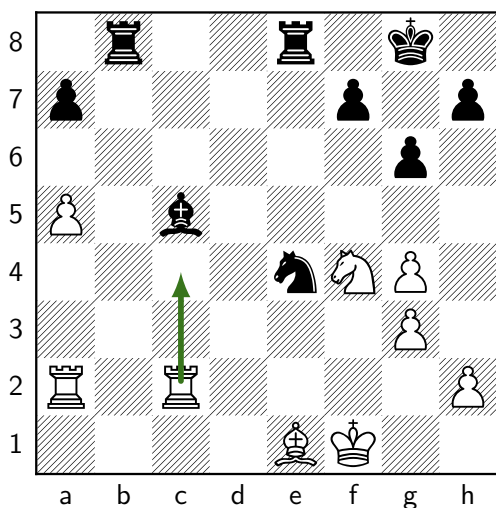
Figure 29: **Move 36, Black: Rb7.** (a) Our position; Melanie plays Rb7 (a7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#578` (ply 75) [structural distance $d=67$, QAP obj=34]: the master played Bh4; under μ our *Ra2* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *Qd6* as their *Rc8*, our *Ra7* as their *Qb7*; transferred move a7b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2834` (ply 64) [structural distance $d=78$, QAP obj=24]: the master played Qg4; under μ our *Ra2* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *Nf6* as their *Bc1*; transferred move a7b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2188` (ply 66) [structural distance $d=77$, QAP obj=13]: the master played Rf1; under μ our *Ra2* plays the role of that game’s *Re1*; cross-type roles: our *Bc8* as their *Rc2*, our *ng5* as their *pc4*; transferred move a7b7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb7 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



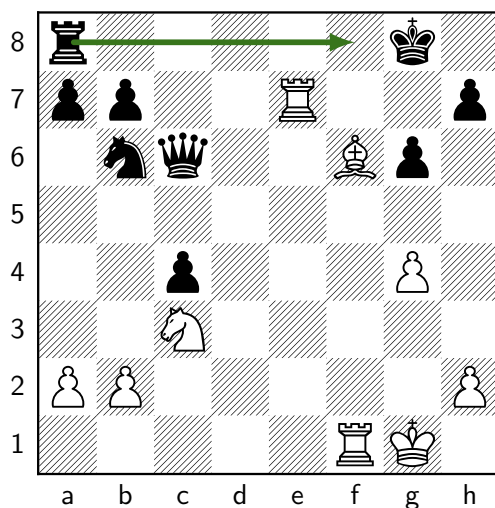
(a)



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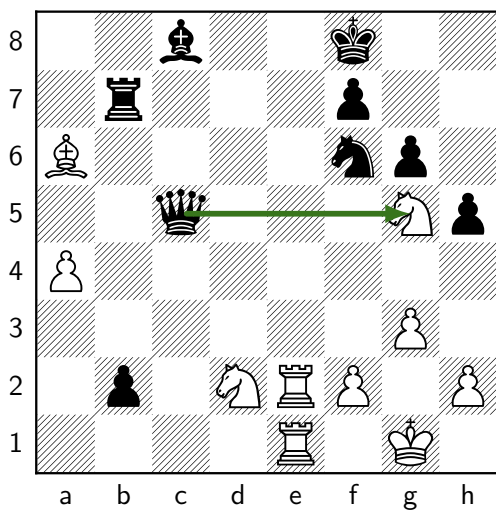


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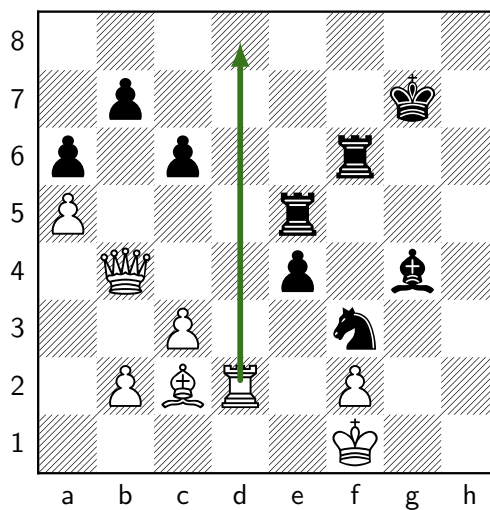


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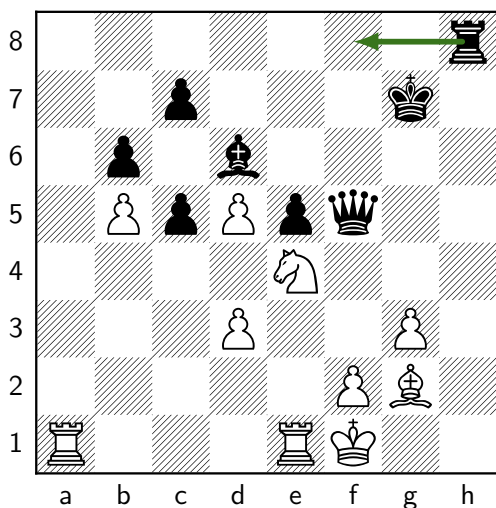
Figure 30: **Move 37, Black: Qc5**. (a) Our position; Melanie plays Qc5 (d6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3433` (ply 72) [structural distance $d=75$, QAP obj=39]: the master played Ne6; under μ our Qd3 plays the role of that game’s Ng5 (a **cross-type** correspondence); cross-type roles: our re2 as their Bb2, our Ph5 as their Re2; transferred move d6c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2494` (ply 60) [structural distance $d=67$, QAP obj=30]: the master played Rc4; under μ our Qd3 plays the role of that game’s Rc2 (a **cross-type** correspondence); cross-type roles: our Rb7 as their Be1, our Nf6 as their Ra2; transferred move d6c5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2715` (ply 57) [structural distance $d=72$, QAP obj=28]: the master played Rf8; under μ our Rb2 plays the role of that game’s Ra1; cross-type roles: our ng5 as their Pa7, our bc4 as their re7; transferred move b7e7. Synthesis: 2 retrieved analogs transferred to Qc5 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



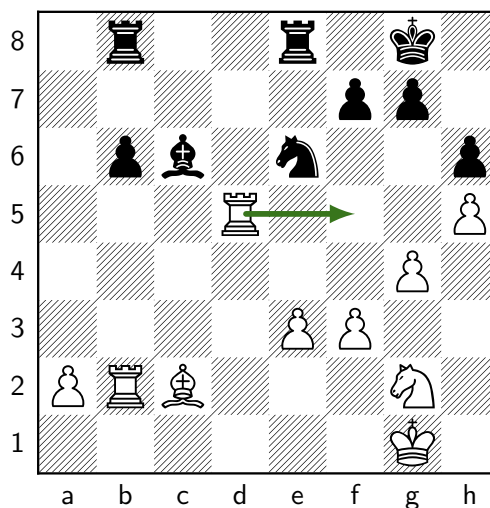
(a)



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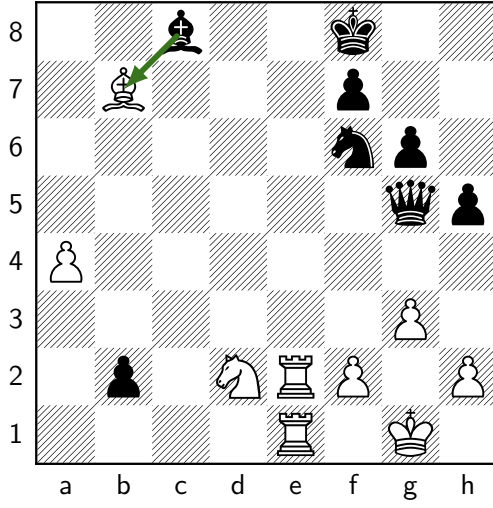


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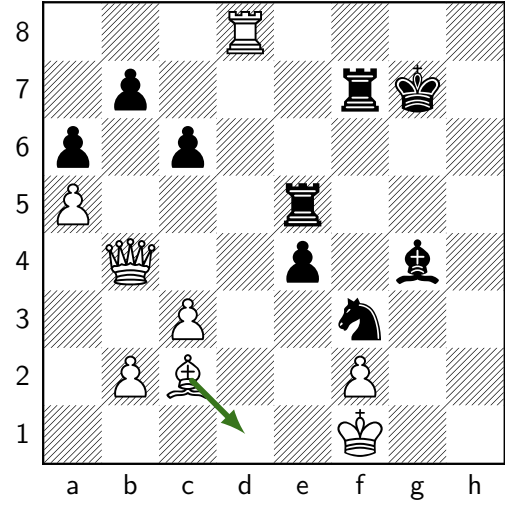


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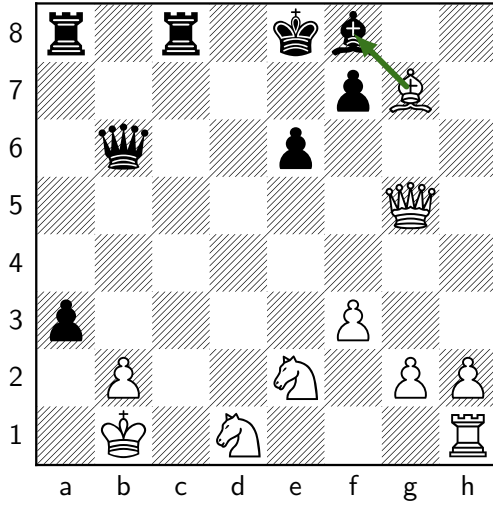
Figure 31: **Move 38, Black: Qxg5.** (a) Our position; Melanie plays Qxg5 (c5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1206` (ply 76) [structural distance $d=64$, QAP obj=42]: the master played Rd8; under μ our Rb2 plays the role of that game’s Rd2; cross-type roles: our Pb2 as their Bc2, our Bc8 as their Pc3; transferred move b7b4. (c) Retrieved from `ccr1_top_p0002#1758` (ply 75) [structural distance $d=76$, QAP obj=40]: the master played Rf8; under μ our Rb2 plays the role of that game’s Rh1; cross-type roles: our Bc8 as their Pb6, our Nf6 as their Bd6; transferred move b7e7. (d) Retrieved from `ccr1_top_p0002#1200` (ply 88) [structural distance $d=72$, QAP obj=36]: the master played Rf5; under μ our Rb2 plays the role of that game’s Rd5; cross-type roles: our Bc8 as their Pa2, our Qc5 as their Rb2; transferred move b7b6. Synthesis: no retrieved analog transferred to Qxg5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



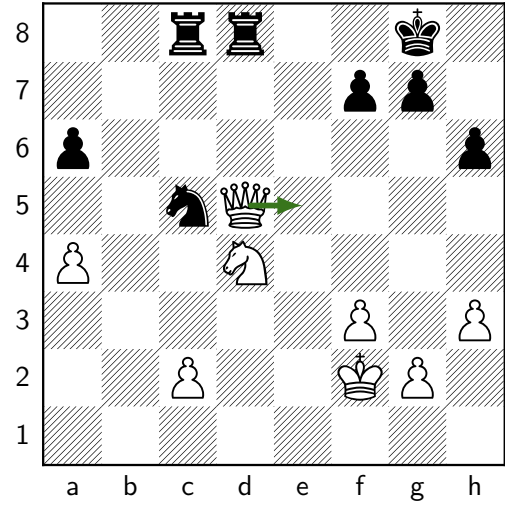
(a)



(b)



(c)



(d)

Figure 32: **Move 39, Black: Bxb7.** (a) Our position; Melanie plays Bxb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1206` (ply 78) [structural distance $d=50$, QAP obj=31]: the master played Bd1; under μ our Bc1 plays the role of that game’s Bc2; cross-type roles: our re2 as their nf3, our re1 as their bg4; transferred move c8b7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2585` (ply 62) [structural distance $d=72$, QAP obj=49]: the master played Bxf8; under μ our Bc1 plays the role of that game’s Bg7; cross-type roles: our pf2 as their Nd1, our re2 as their Ne2; transferred move c8h3. (d) Retrieved from `ccrl_top_p0001#864` (ply 68) [structural distance $d=56$, QAP obj=47]: the master played Qe5; under μ our Qg4 plays the role of that game’s Qd5; cross-type roles: our Bc8 as their Ph3, our bb7 as their Pa4; transferred move g5e5. Synthesis: 1 retrieved analog transferred to Bxb7 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

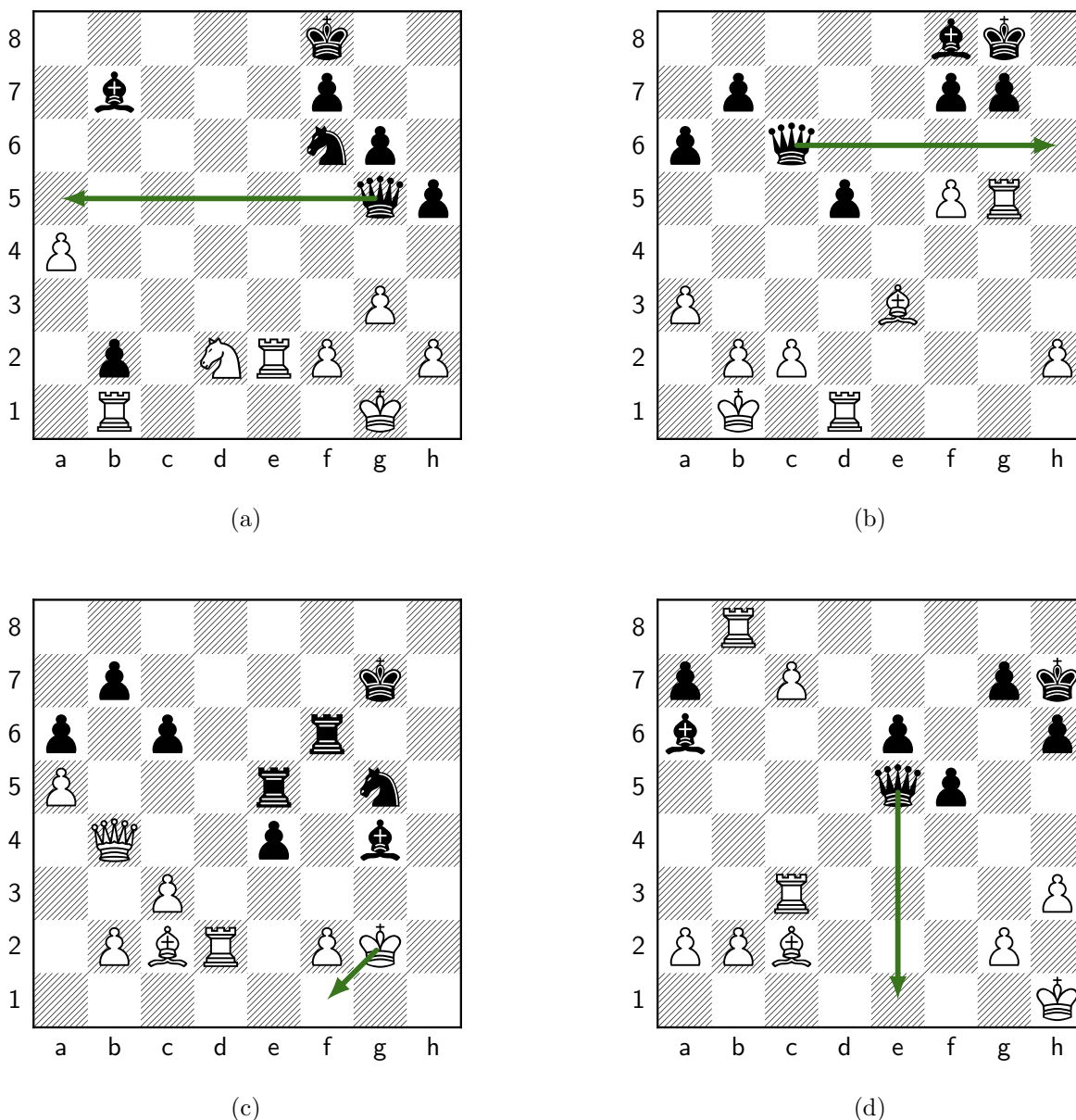
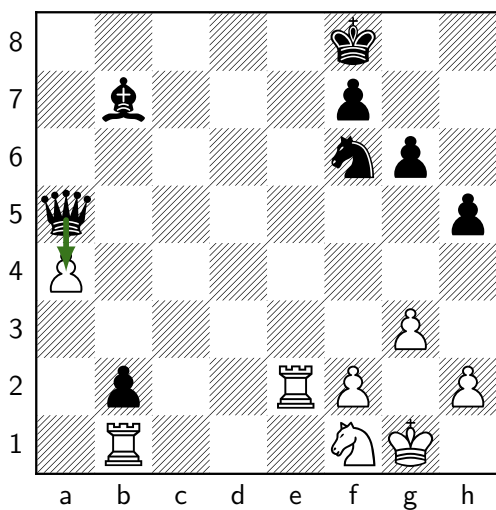
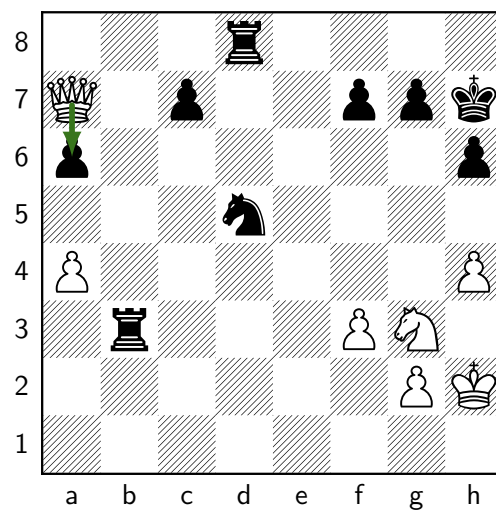


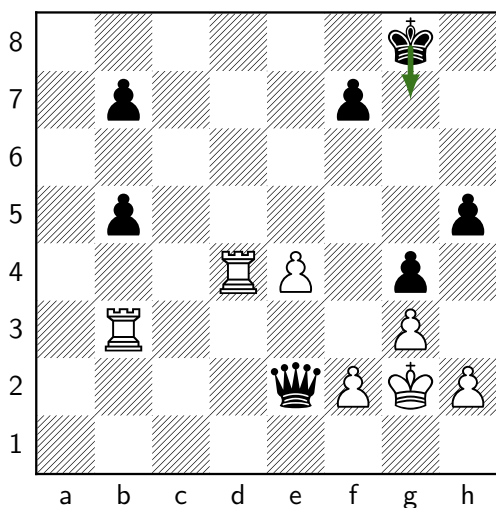
Figure 33: **Move 40, Black: Qa5.** (a) Our position; Melanie plays Qa5 (g5a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2947` (ply 65) [structural distance $d=57$, QAP obj=30]: the master played Qh6; under μ our Qg4 plays the role of that game’s Qc3; cross-type roles: our re2 as their Bf8, our Nf6 as their Pa6; transferred move g5a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1206` (ply 74) [structural distance $d=52$, QAP obj=42]: the master played Kf1; under μ our Kf1 plays the role of that game’s Kg2; cross-type roles: our Qg5 as their Bc2, our Nf6 as their Rd2; transferred move f8g7. (d) Retrieved from `ccrl_top_p0001#2288` (ply 63) [structural distance $d=55$, QAP obj=41]: the master played Qe1+; under μ our Qg4 plays the role of that game’s Qe4; cross-type roles: our Nf6 as their Ba6, our nd2 as their rc3; transferred move g5d2. Synthesis: 1 retrieved analog transferred to Qa5 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



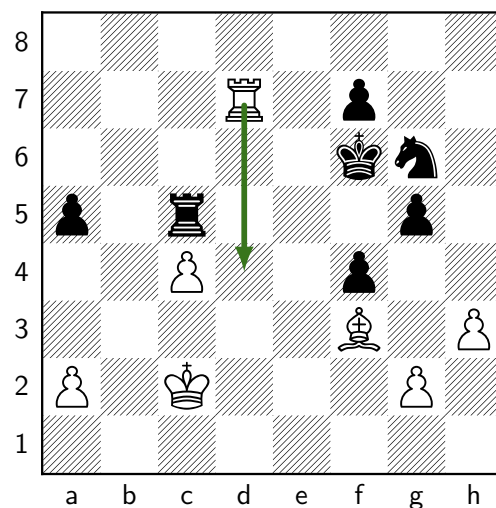
(a)



(b)

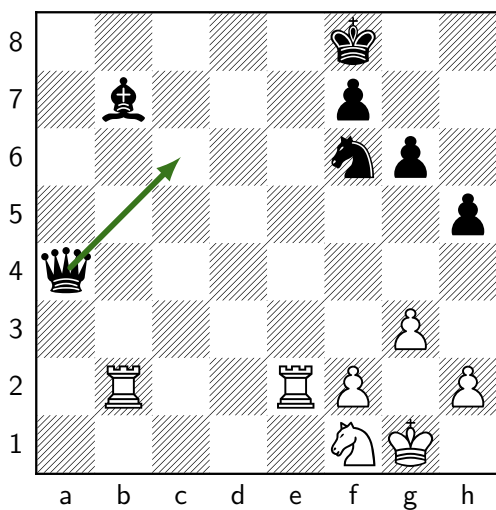


(c)

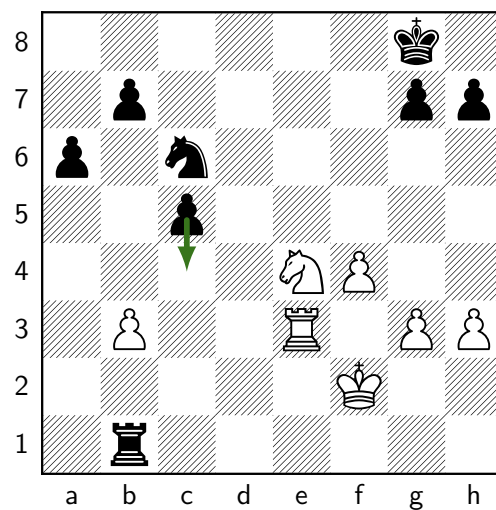


(d)

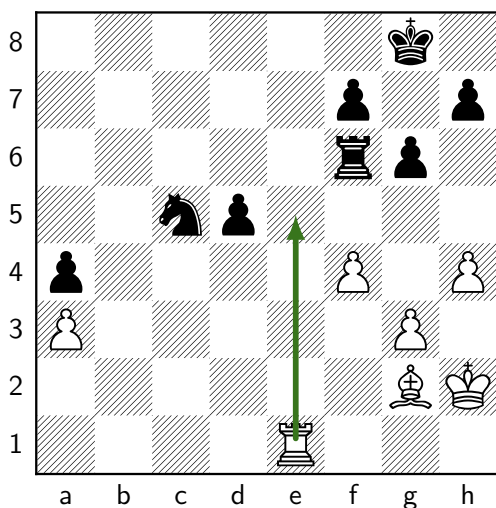
Figure 34: **Move 41, Black: Qxa4.** (a) Our position; Melanie plays Qxa4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2796` (ply 70) [structural distance $d=55$, QAP obj=36]: the master played Qxa6; under μ our Qa4 plays the role of that game’s Qa7; cross-type roles: our Bb7 as their Pa4, our pf2 as their nd5; transferred move a5a4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#97` (ply 69) [structural distance $d=53$, QAP obj=57]: the master played Kg7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Bb7 as their Pb7, our pf2 as their rd4; transferred move f8g7. (d) Retrieved from `ccrl_top_p0002#884` (ply 76) [structural distance $d=53$, QAP obj=52]: the master played Rd4; under μ our Qa4 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pa2, our Nf6 as their Bf3; transferred move a5c5. Synthesis: 1 retrieved analog transferred to Qxa4 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



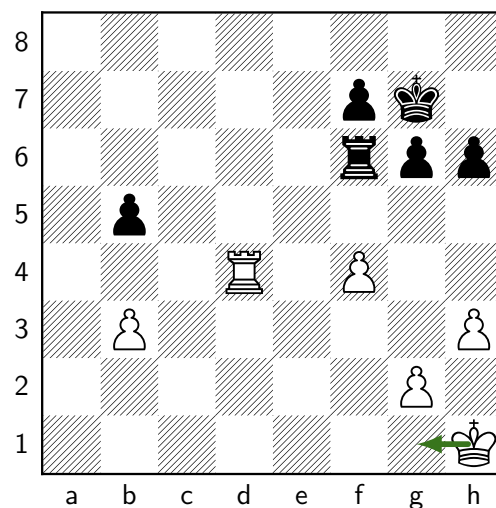
(a)



(b)



(c)



(d)

Figure 35: **Move 42, Black: Qc6.** (a) Our position; Melanie plays Qc6 (a4c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2142` (ply 75) [structural distance $d=49$, QAP obj=22]: the master played c4; under μ our *Bb2* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *Ph5* as their *Nc6*, our *rb2* as their *ne4*; transferred move b7f3. (c) Retrieved from `ccrl_top_p0000#2893` (ply 80) [structural distance $d=49$, QAP obj=19]: the master played Re5; under μ our *Qa5* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *rb2* as their *Bg2*, our *Nf6* as their *Pg3*; transferred move a4f4. (d) Retrieved from `ccrl_top_p0003#1097` (ply 72) [structural distance $d=50$, QAP obj=18]: the master played Kg1; under μ our *Kf1* plays the role of that game’s *Kh1*; cross-type roles: our *Nf6* as their *Ph3*, our *Qa4* as their *Rd4*; transferred move f8g7. Synthesis: no retrieved analog transferred to Qc6 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

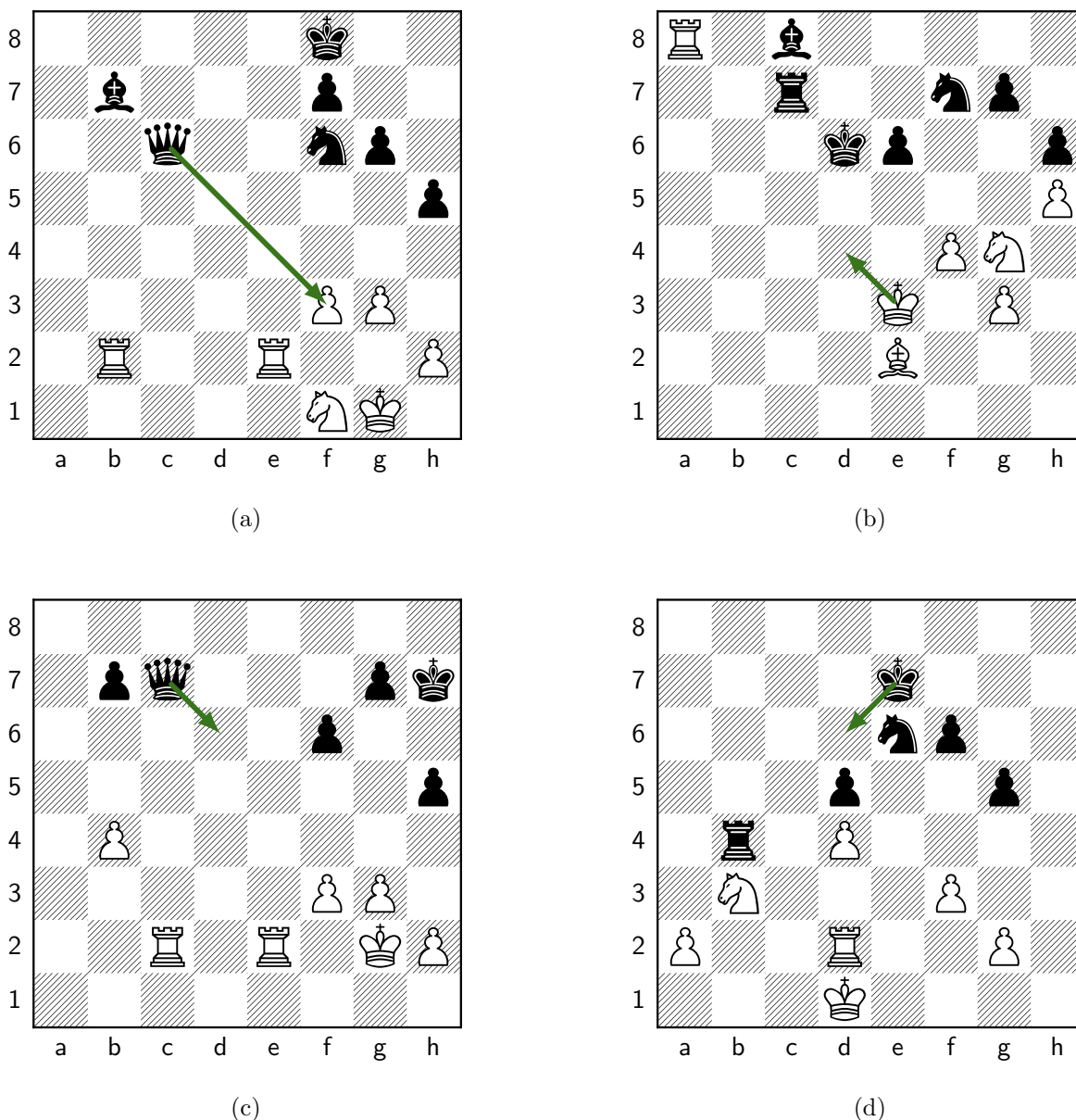
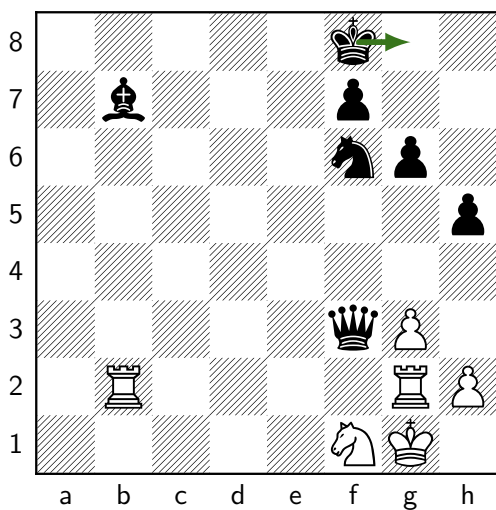
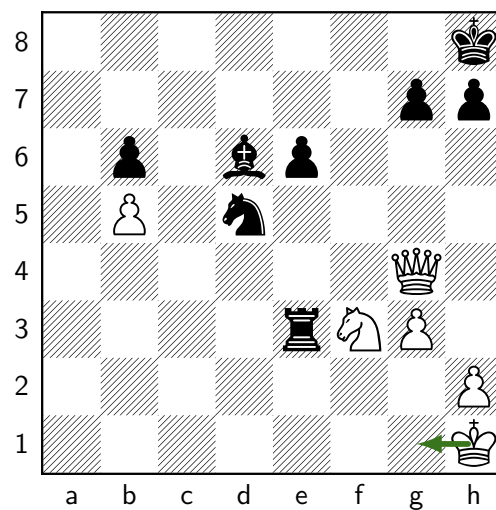


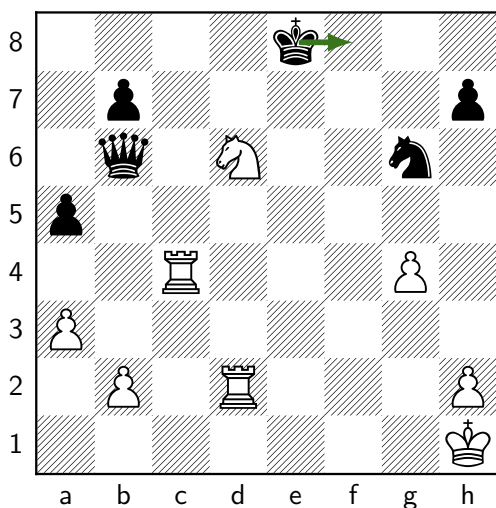
Figure 36: **Move 43, Black: Qxf3**. (a) Our position; Melanie plays **Qxf3** (c6f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#954` (ply 96) [structural distance $d=54$, QAP obj=14]: the master played **Kd4**; under μ our **Kf1** plays the role of that game's **Ke3**; cross-type roles: our **nf1** as their **Be2**, our **Nf6** as their **Pf4**; transferred move **f8g8**. (c) Retrieved from `ccr1_top_p0002#2180` (ply 83) [structural distance $d=52$, QAP obj=10]: the master played **Qd6**; under μ our **Bb2** plays the role of that game's **Qc2** (a **cross-type** correspondence); cross-type roles: our **Qc6** as their **Pb7**, our **Nf6** as their **Pf6**; transferred move **b7a8**. (d) Retrieved from `ccr1_top_p0002#2394` (ply 85) [structural distance $d=51$, QAP obj=5]: the master played **Kd6**; under μ our **Kf1** plays the role of that game's **Ke2**; cross-type roles: our **re2** as their **Ne6**, our **Nf6** as their **Pd5**; transferred move **f8g7**. Synthesis: no retrieved analog transferred to **Qxf3** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



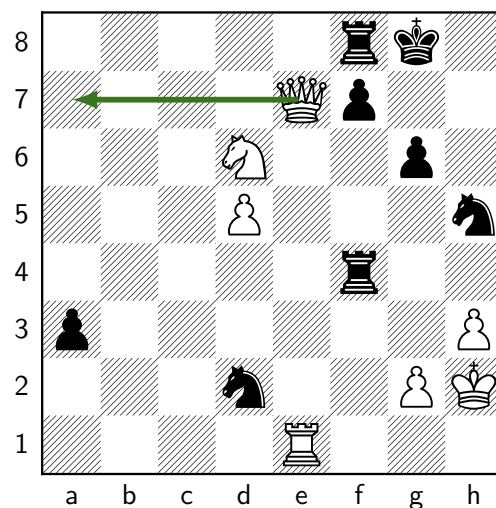
(a)



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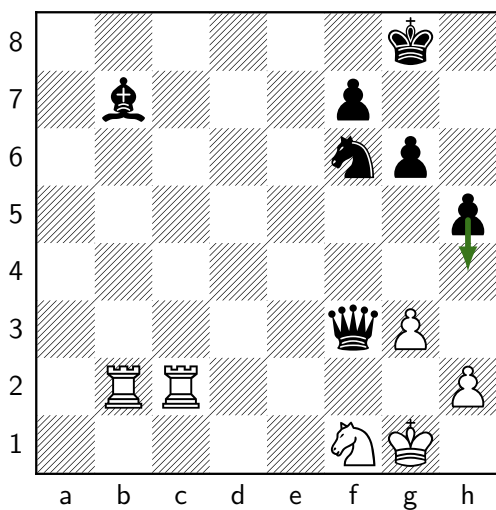


(c)

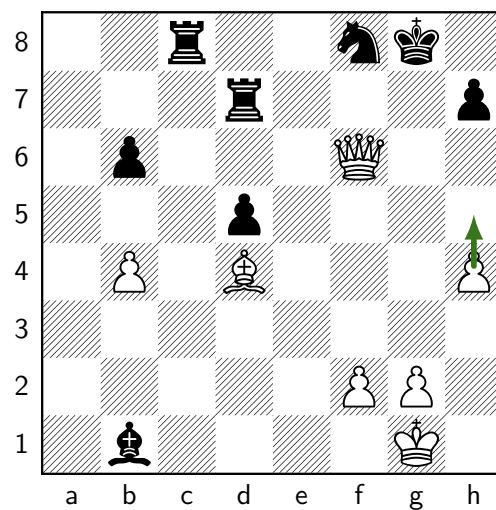


(d)

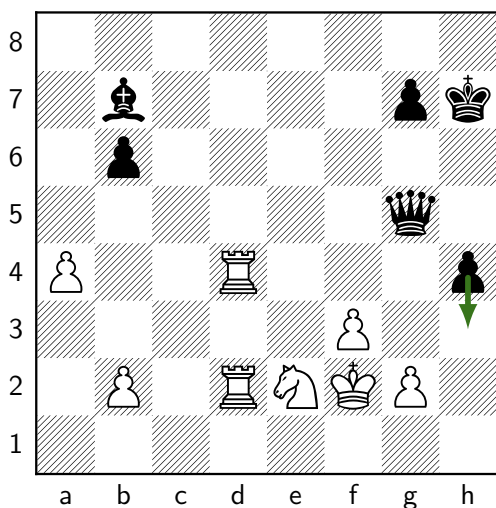
Figure 37: **Move 44, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3662` (ply 74) [structural distance $d=80$, QAP obj=15]: the master played Kg1; under μ our Kf1 plays the role of that game’s Kh1; cross-type roles: our Qf3 as their re3, our rg2 as their Nf3; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2715` (ply 95) [structural distance $d=65$, QAP obj=2]: the master played Kf8; under μ our Kf1 plays the role of that game’s Ke1; cross-type roles: our Bb7 as their Pb7, our pg3 as their nd6; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3792` (ply 86) [structural distance $d=80$, QAP obj=38]: the master played Qa7; under μ our Qf6 plays the role of that game’s Qe7; cross-type roles: our Bb7 as their Re1, our Pf7 as their nd2; transferred move f3a3. Synthesis: 2 retrieved analogs transferred to Kg8 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



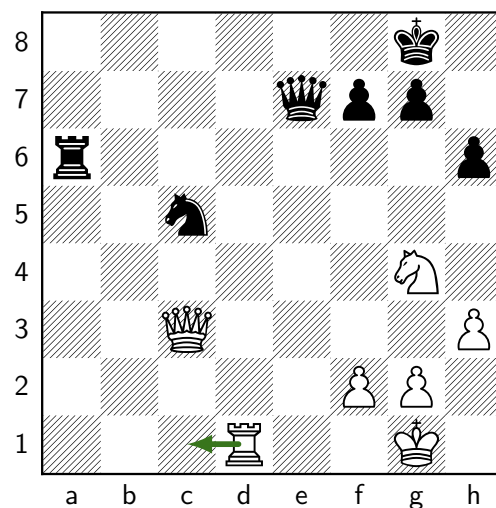
(a)



(b)

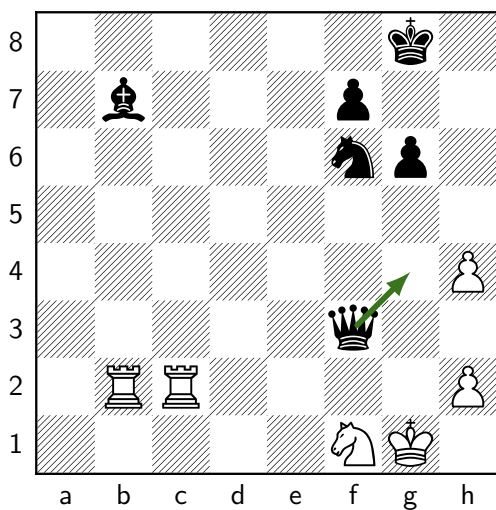


(c)

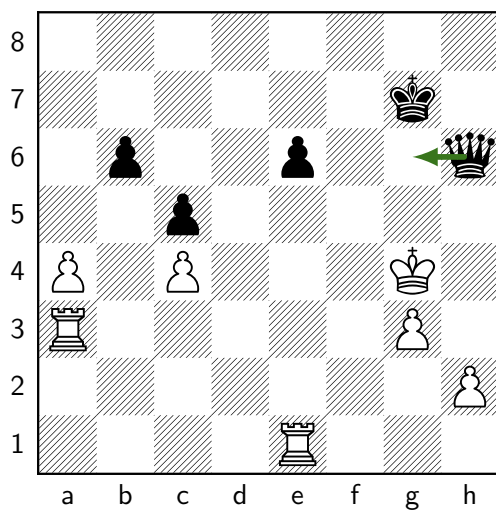


(d)

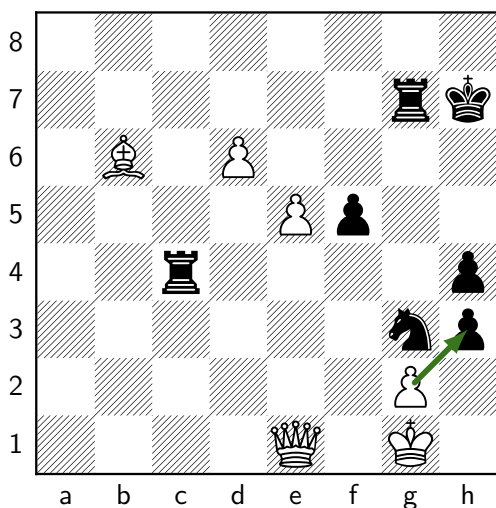
Figure 38: **Move 45, Black: h4 (h5h4).** (a) Our position; Melanie plays h4 (h5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#866` (ply 74) [structural distance $d=50$, QAP obj=30]: the master played h5; under μ our Ph_4 plays the role of that game’s Ph_4 ; cross-type roles: our Pf_7 as their bb_1 , our Nf_6 as their Pf_2 ; transferred move h5h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3040` (ply 99) [structural distance $d=69$, QAP obj=24]: the master played h3; under μ our Ph_4 plays the role of that game’s Ph_5 ; cross-type roles: our Bb_7 as their Pg_7 , our rb_2 as their pa_4 ; transferred move h5h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2484` (ply 80) [structural distance $d=66$, QAP obj=29]: the master played Rc1; under μ our Nf_3 plays the role of that game’s Rd_1 (a **cross-type** correspondence); cross-type roles: our rc_2 as their Ph_3 , our rb_2 as their Ng_4 ; transferred move f6e8. Synthesis: 2 retrieved analogs transferred to h4 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



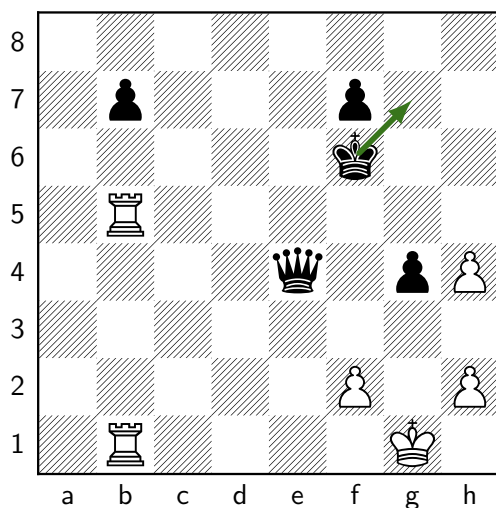
(a)



(b)

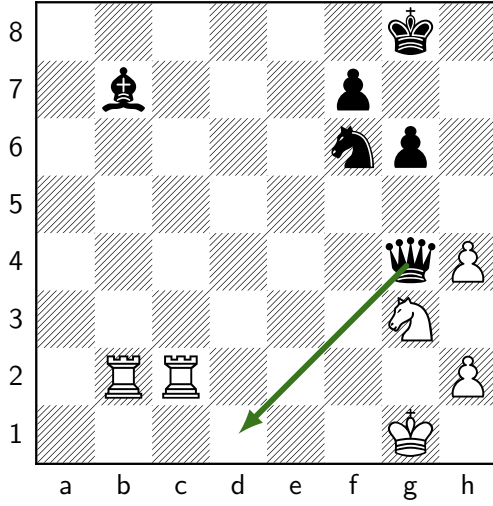


(c)

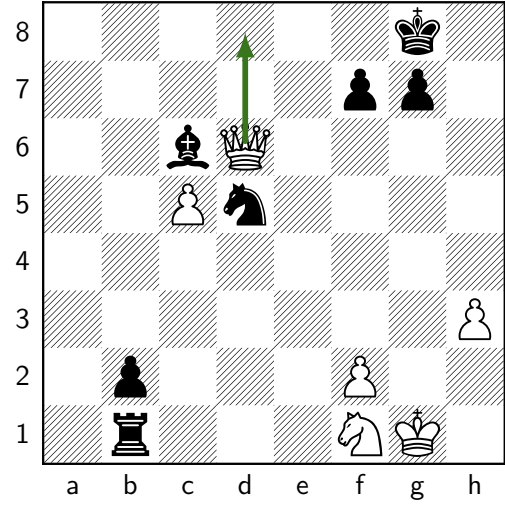


(d)

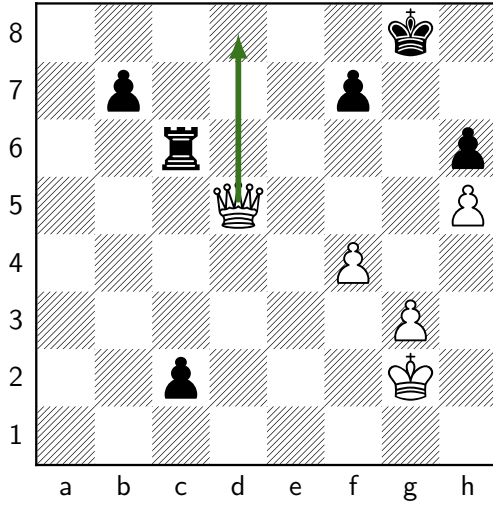
Figure 39: **Move 46, Black: Qg4+ (f3g4).** (a) Our position; Melanie plays Qg4+ (f3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#2473` (ply 87) [structural distance $d=52$, QAP obj=4]: the master played Qg6+; under μ our Qf6 plays the role of that game’s Qh3; cross-type roles: our Bb7 as their Pe6, our Nf6 as their Pc5; transferred move f3g4 (**endorses this move**). (c) Retrieved from `ccl1_top_p0003#2170` (ply 100) [structural distance $d=58$, QAP obj=27]: the master played gxh3; under μ our Bb2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our rb2 as their ph3, our ph4 as their rc4; transferred move b7a6. (d) Retrieved from `ccl1_top_p0003#97` (ply 91) [structural distance $d=45$, QAP obj=24]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kf3; cross-type roles: our Bb7 as their Pb7, our Nf6 as their Pf7; transferred move g8g7. Synthesis: 1 retrieved analog transferred to Qg4+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



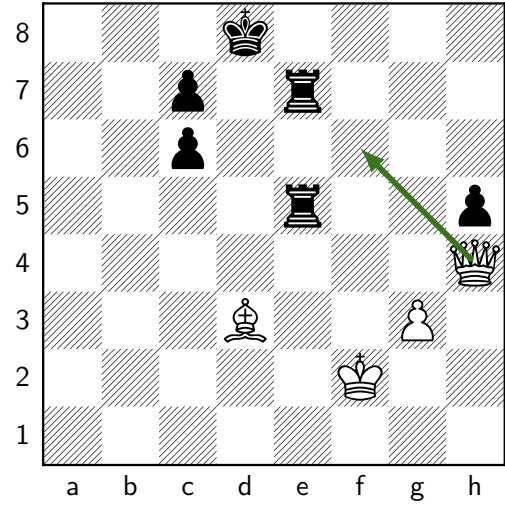
(a)



(b)

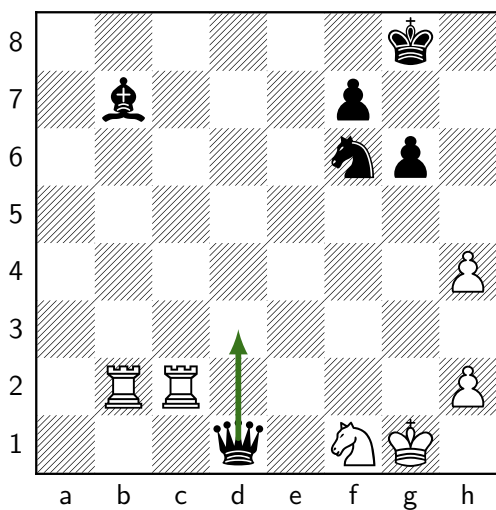


(c)

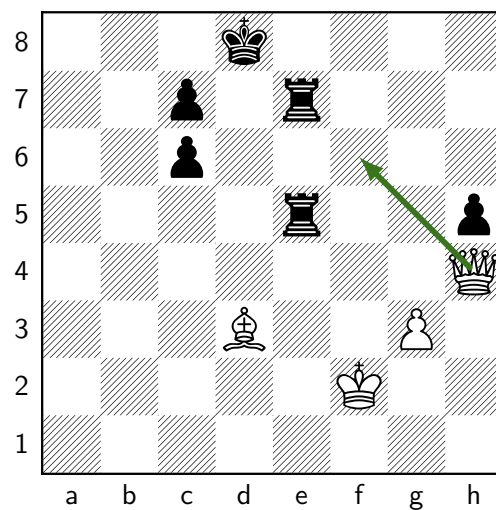


(d)

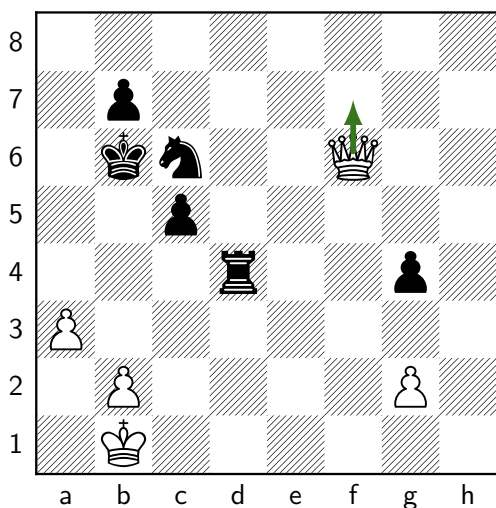
Figure 40: **Move 47, Black: Qd1+.** (a) Our position; Melanie plays Qd1+ (g4d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2956` (ply 98) [structural distance $d=57$, QAP obj=38]: the master played Qd8+; under μ our Qg5 plays the role of that game’s Qd6; cross-type roles: our Qg4 as their rb1, our Bb7 as their Pf2; transferred move g4d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2993` (ply 102) [structural distance $d=57$, QAP obj=17]: the master played Qd8+; under μ our Qg5 plays the role of that game’s Qd5; cross-type roles: our Bb7 as their Pg3, our rc2 as their Pf4; transferred move g4d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#187` (ply 88) [structural distance $d=51$, QAP obj=39]: the master played Qf6; under μ our Qg5 plays the role of that game’s Qh4; cross-type roles: our Nf6 as their Bd3, our ph2 as their re5; transferred move g4d4. Synthesis: 2 retrieved analogs transferred to Qd1+ (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



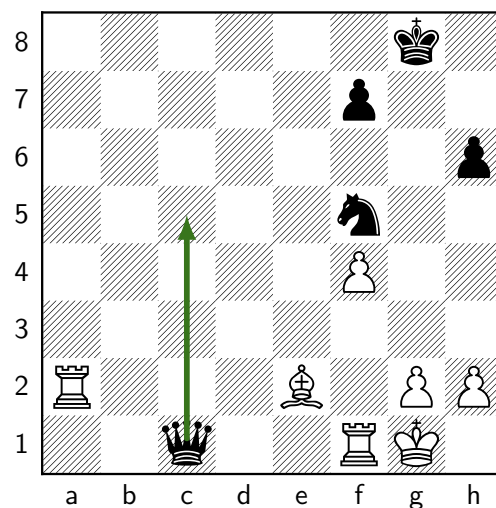
(a)



(b)

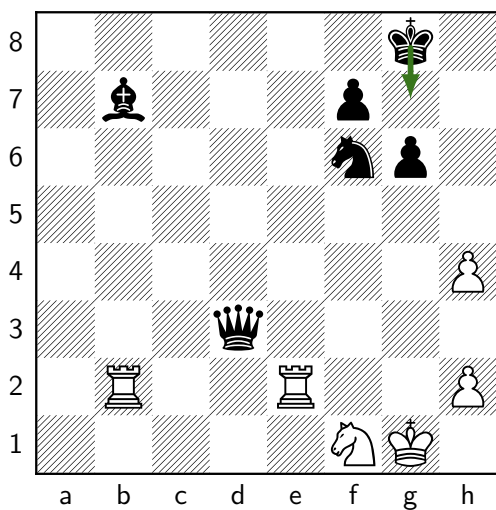


(c)

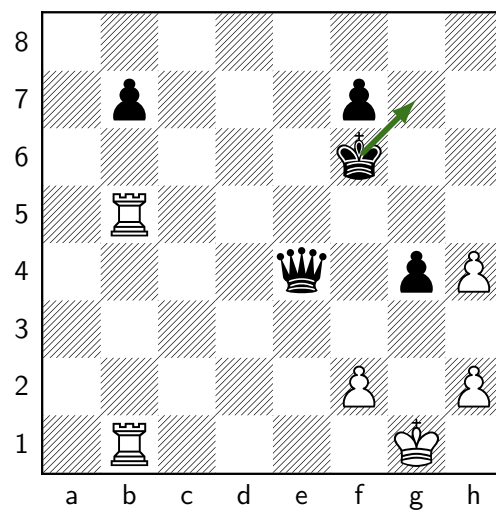


(d)

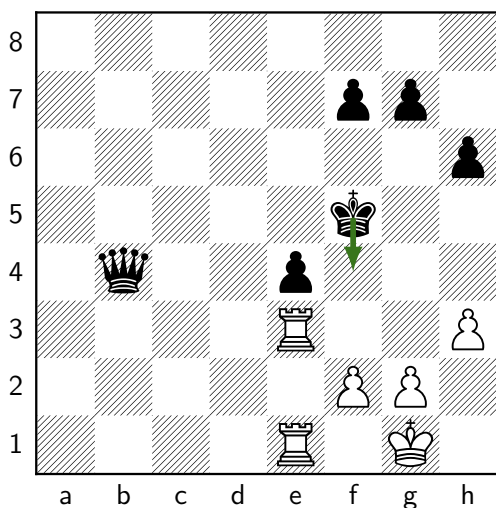
Figure 41: **Move 48, Black: Qd3.** (a) Our position; Melanie plays Qd3 (d1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#187` (ply 88) [structural distance $d=34$, QAP obj=29]: the master played Qf6; under μ our Qd8 plays the role of that game’s Qh4; cross-type roles: our ph4 as their Bd3, our rc2 as their Pg3; transferred move d1d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2777` (ply 80) [structural distance $d=44$, QAP obj=37]: the master played Qf7; under μ our Qd8 plays the role of that game’s Qf6; cross-type roles: our Nf6 as their Pa3, our rb2 as their pc5; transferred move d1e1. (d) Retrieved from `ccr1_top_p0002#1595` (ply 81) [structural distance $d=31$, QAP obj=29]: the master played Qc5+; under μ our Qd8 plays the role of that game’s Qc8; cross-type roles: our Bb7 as their Pf7, our rb2 as their be2; transferred move d1d6. Synthesis: 1 retrieved analog transferred to Qd3 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



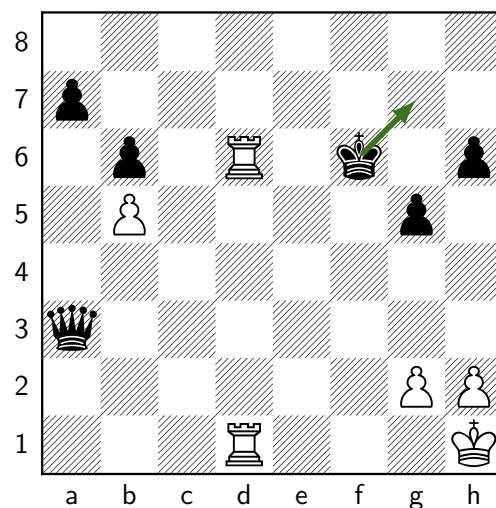
(a)



(b)



(c)



(d)

Figure 42: **Move 49, Black: Kg7.** (a) Our position; Melanie plays Kg7 (**g8g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#97` (ply 91) [structural distance $d=29$, QAP obj=11]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kf3; cross-type roles: our Bb7 as their Pb7, our Nf6 as their Pf7; transferred move **g8g7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#2691` (ply 97) [structural distance $d=42$, QAP obj=10]: the master played Kf4; under μ our Kg1 plays the role of that game’s Kf4; cross-type roles: our Nf6 as their Pf7, our Bb7 as their Pe4; transferred move **g8g7 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#1185` (ply 93) [structural distance $d=40$, QAP obj=4]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kf3; cross-type roles: our Nf6 as their Pb6, our Bb7 as their rd6; transferred move **g8g7 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to Kg7 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

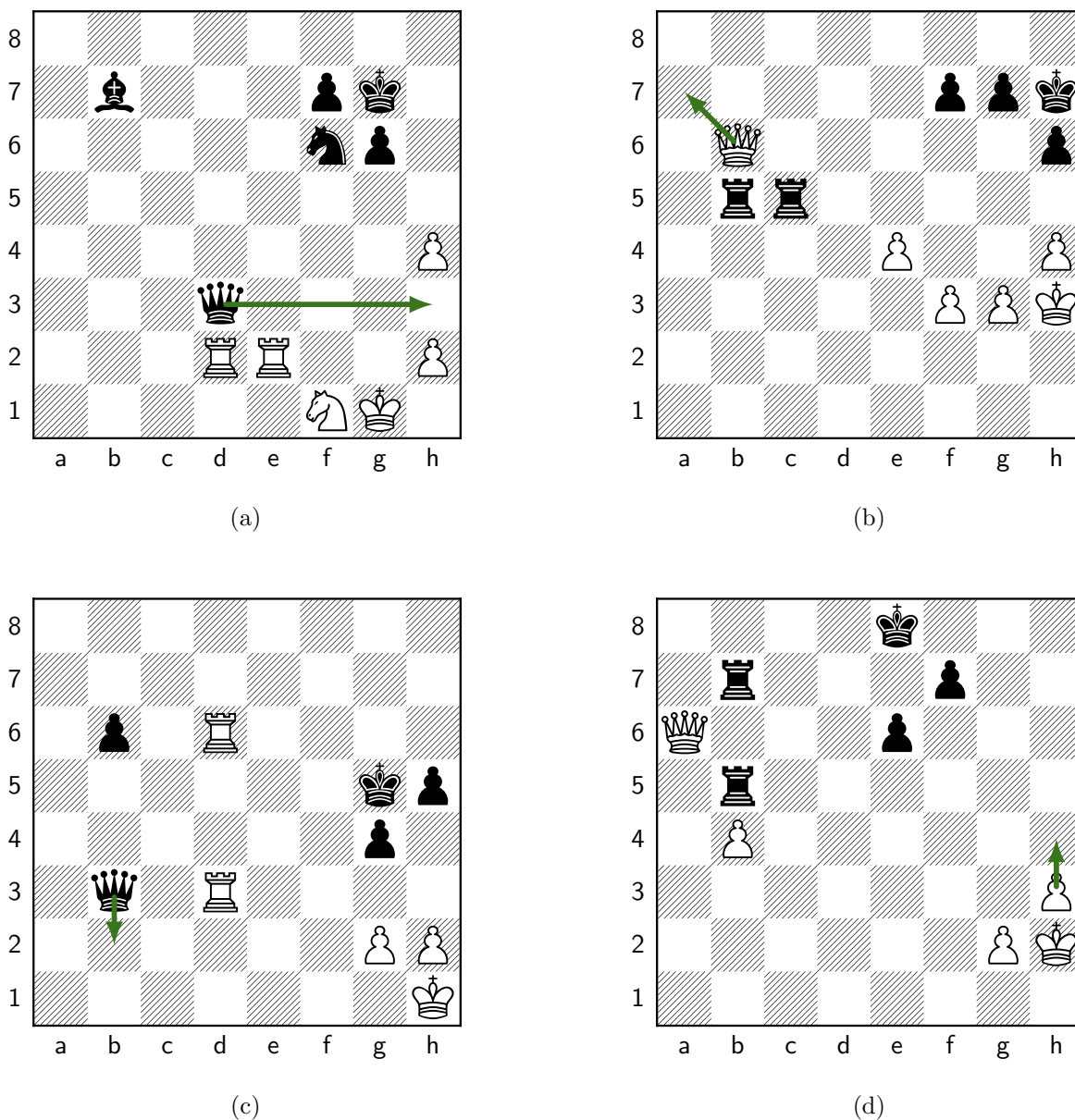
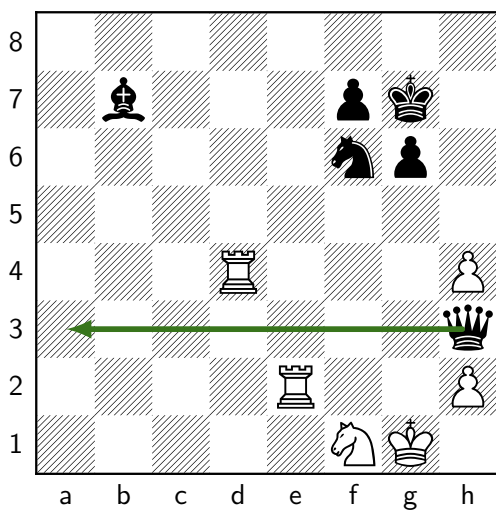
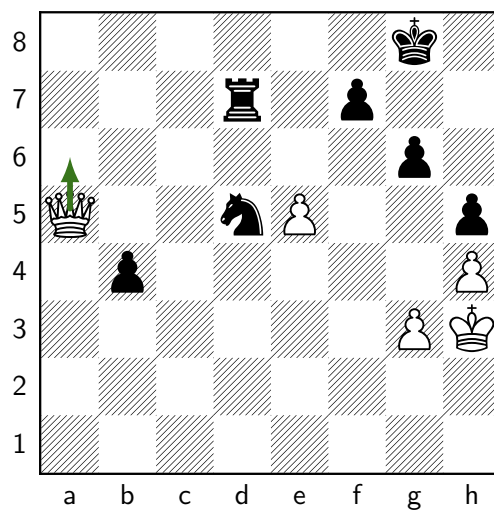


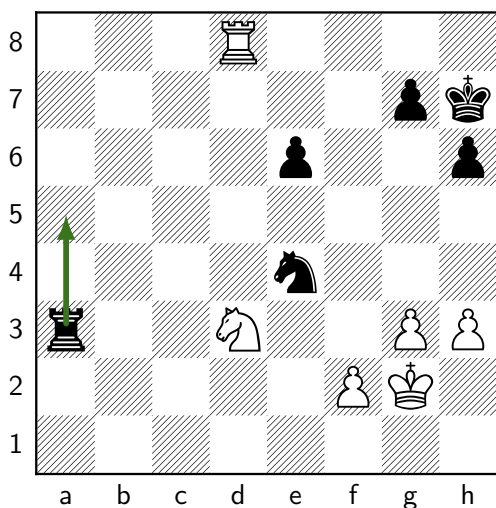
Figure 43: **Move 50, Black: Qh3.** (a) Our position; Melanie plays Qh3 (d3h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3923` (ply 88) [structural distance $d=60$, QAP obj=73]: the master played Qa7; under μ our Qd6 plays the role of that game’s Qb6; cross-type roles: our Bb7 as their Pf3, our Nf6 as their Pe4; transferred move d3h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1185` (ply 111) [structural distance $d=52$, QAP obj=19]: the master played Qb2; under μ our Qd6 plays the role of that game’s Qb6; cross-type roles: our Nf6 as their Pb6, our Bb7 as their pg2; transferred move d3h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2590` (ply 102) [structural distance $d=56$, QAP obj=34]: the master played h4; under μ our Nf3 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pb4, our nf1 as their pf7; transferred move f6e8. Synthesis: 2 retrieved analogs transferred to Qh3 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



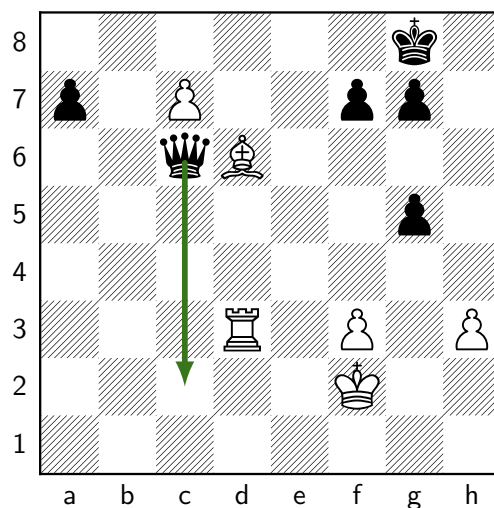
(a)



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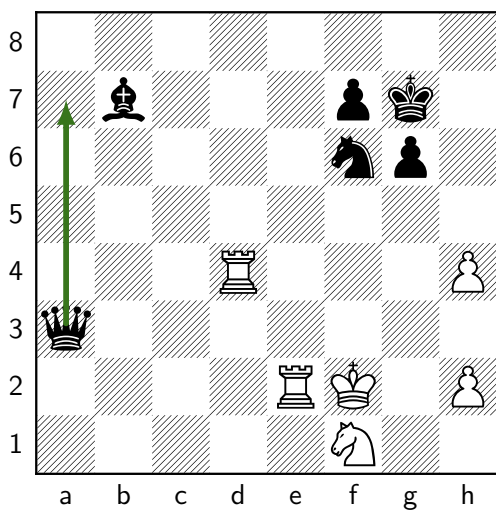


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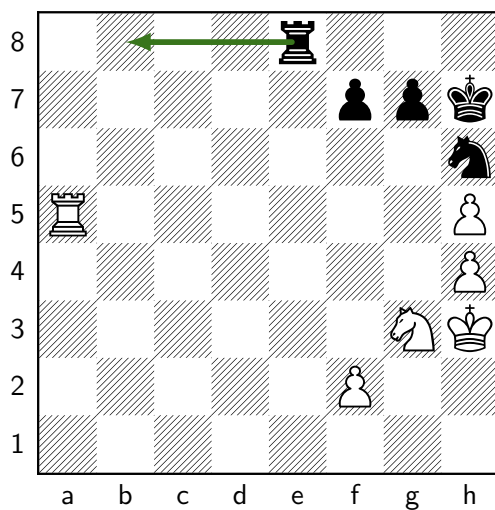


(d)

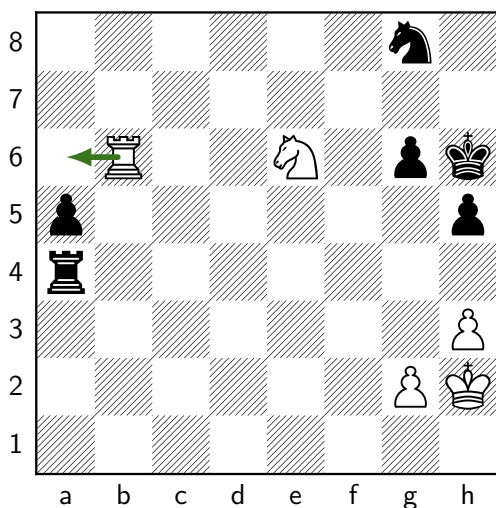
Figure 44: **Move 51, Black: Qa3**. (a) Our position; Melanie plays Qa3 (h3a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2057 (ply 112) [structural distance $d=53$, QAP obj=35]: the master played Qa6; under μ our Qh6 plays the role of that game’s Qa5; cross-type roles: our Nf6 as their Pe5, our Bb7 as their pg6; transferred move h3a3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#163 (ply 117) [structural distance $d=52$, QAP obj=27]: the master played Ra5; under μ our Qh6 plays the role of that game’s Ra6 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Pe6, our Qh3 as their Ne4; transferred move h3a3 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#853 (ply 115) [structural distance $d=47$, QAP obj=27]: the master played Qc2+; under μ our Qh6 plays the role of that game’s Qc3; cross-type roles: our Bb7 as their Pa7, our nf1 as their bd6; transferred move h3a3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qa3 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



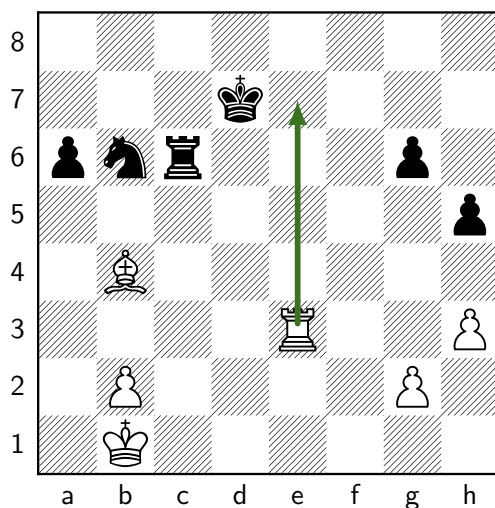
(a)



(b)

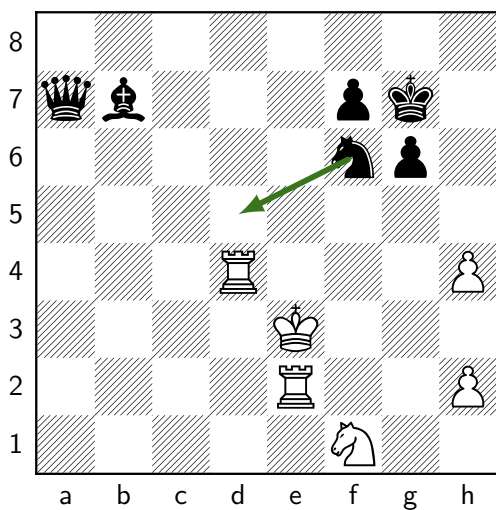


(c)

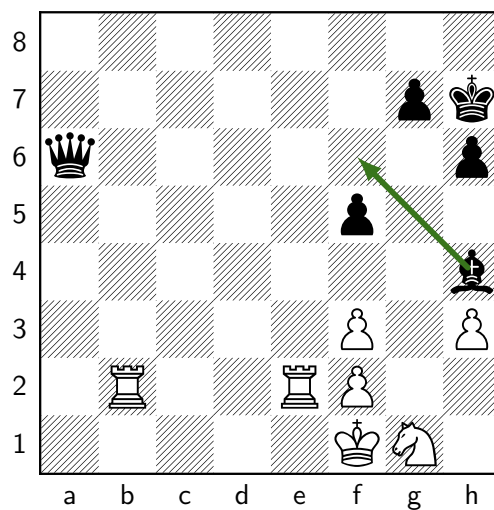


(d)

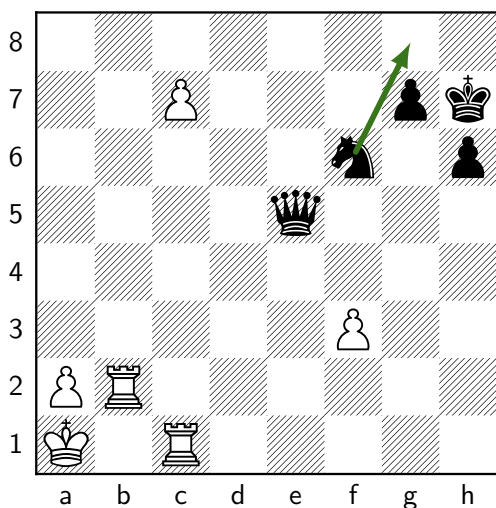
Figure 45: **Move 52, Black: Qa7**. (a) Our position; Melanie plays Qa7 (a3a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#660` (ply 95) [structural distance $d=42$, QAP obj=1]: the master played Rb8; under μ our Qa6 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our rd4 as their Pf7, our re2 as their Nh6; transferred move a3a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2665` (ply 88) [structural distance $d=32$, QAP obj=2]: the master played Ra6; under μ our Qa6 plays the role of that game’s Rb6 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Ph3, our Bb7 as their Ne6; transferred move a3b3. (d) Retrieved from `ccr1_top_p0001#1589` (ply 92) [structural distance $d=42$, QAP obj=2]: the master played Re7+; under μ our Bb2 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Ph3, our Qa3 as their Bb4; transferred move b7g2. Synthesis: 1 retrieved analog transferred to Qa7 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



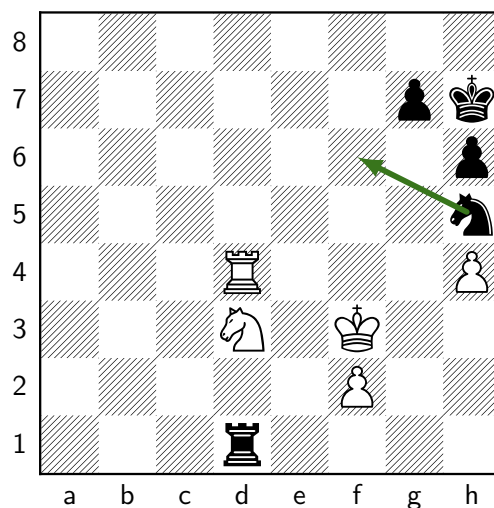
(a)



(b)

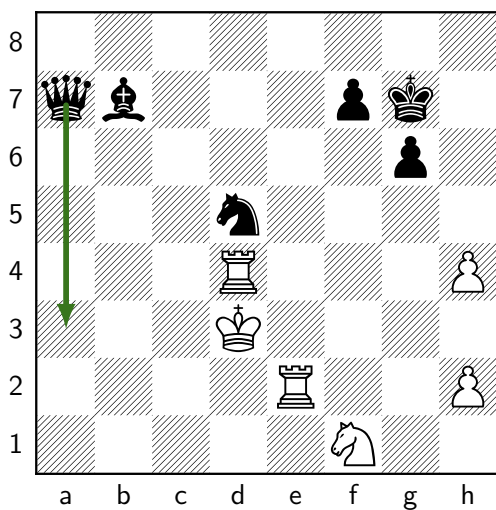


(c)

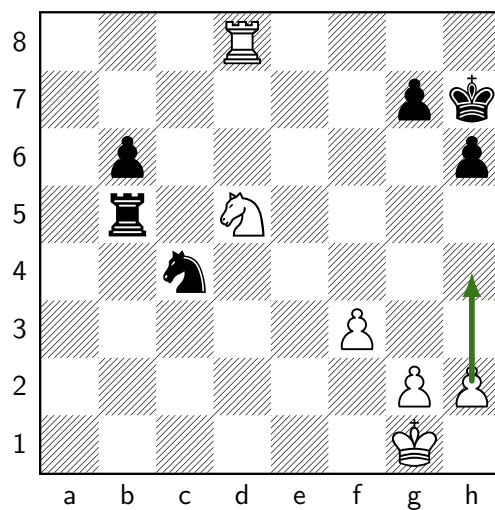


(d)

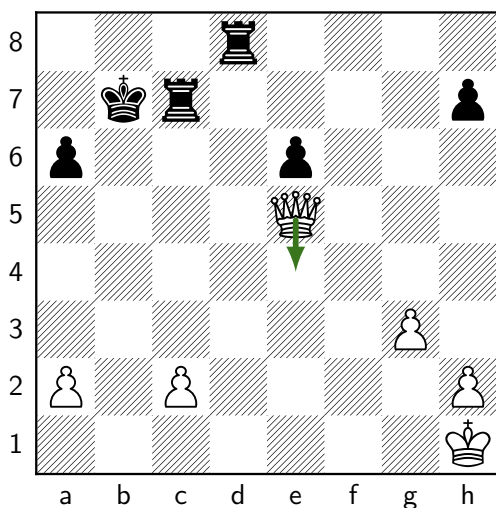
Figure 46: **Move 53, Black: Nd5+.** (a) Our position; Melanie plays Nd5+ (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3845` (ply 107) [structural distance $d=53$, QAP obj=34]: the master played Bf6; under μ our Nf3 plays the role of that game’s Bh5 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Ph6, our nf1 as their Pf5; transferred move f6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1424` (ply 117) [structural distance $d=37$, QAP obj=48]: the master played Ng8; under μ our Bb2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our rd4 as their pc7, our Nf6 as their Ph6; transferred move b7c8. (d) Retrieved from `ccr1_top_p0001#2604` (ply 91) [structural distance $d=51$, QAP obj=38]: the master played Nf6; under μ our Nf3 plays the role of that game’s Nh4; cross-type roles: our Bb7 as their rd4, our rd4 as their nd3; transferred move f6h5. Synthesis: 1 retrieved analog transferred to Nd5+ (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



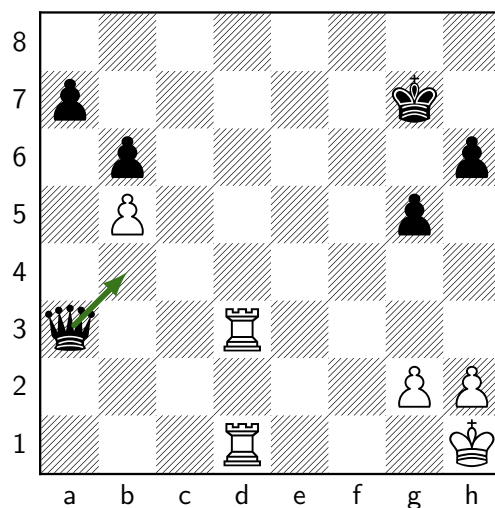
(a)



(b)

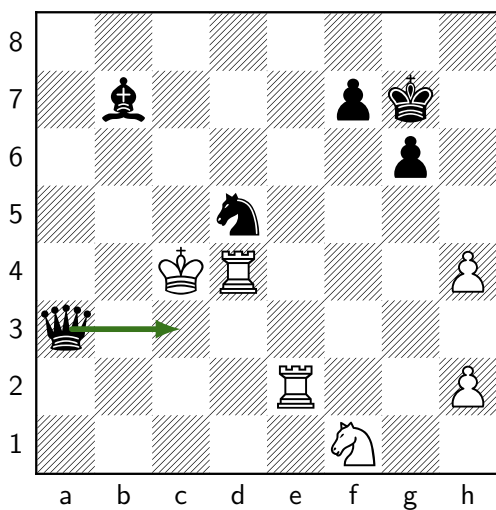


(c)

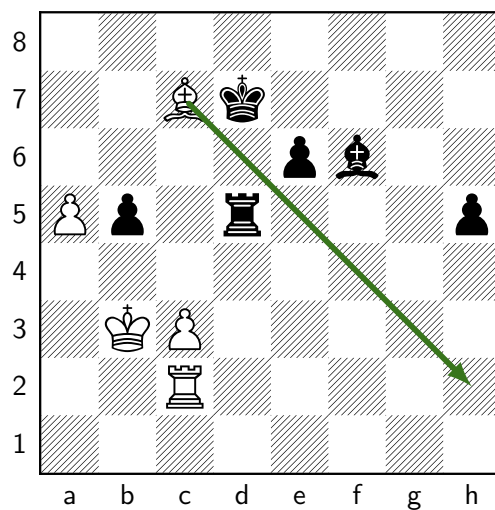


(d)

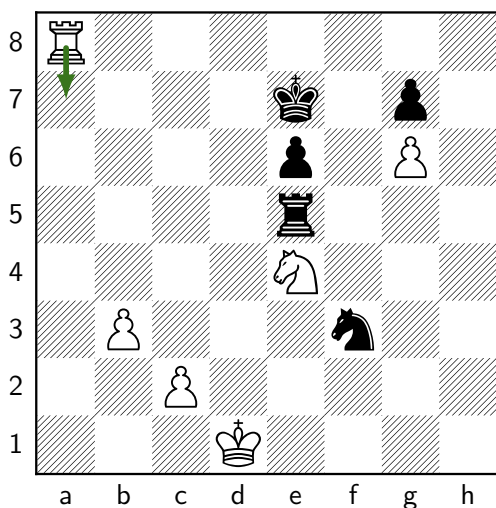
Figure 47: **Move 54, Black: Qa3+.** (a) Our position; Melanie plays Qa3+ (a7a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3506` (ply 104) [structural distance $d=42$, QAP obj=27]: the master played h4; under μ our Pf2 plays the role of that game’s Ph2; cross-type roles: our Bb7 as their nc4, our Qa7 as their rb5; transferred move f7f5. (c) Retrieved from `ccr1_top_p0000#3967` (ply 90) [structural distance $d=44$, QAP obj=22]: the master played Qe4+; under μ our Qa2 plays the role of that game’s Qe5; cross-type roles: our Nd5 as their Pc2, our Bb7 as their Pg3; transferred move a7c5. (d) Retrieved from `ccr1_top_p0000#1185` (ply 95) [structural distance $d=37$, QAP obj=20]: the master played Qb4; under μ our Nd4 plays the role of that game’s Qa6 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pa7, our re2 as their Pb6; transferred move d5c7. Synthesis: no retrieved analog transferred to Qa3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



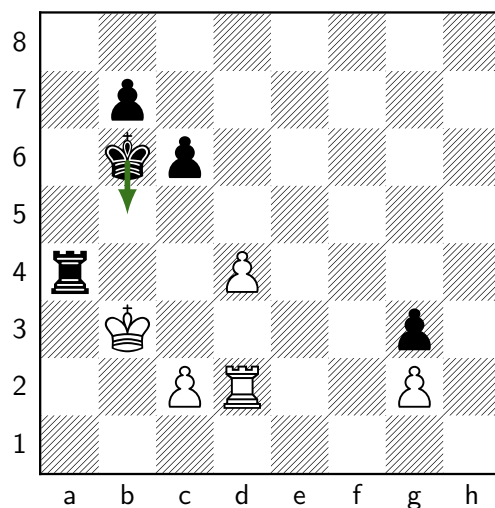
(a)



(b)

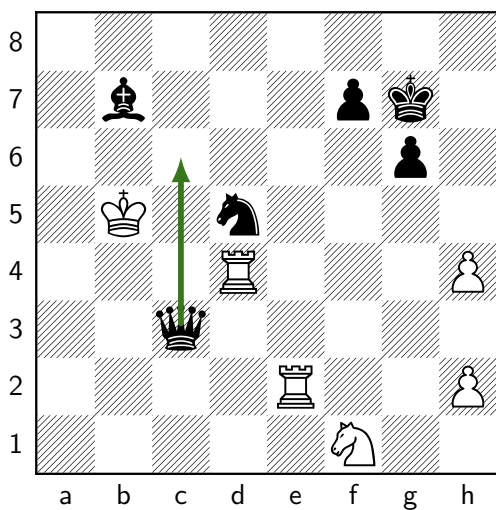


(c)

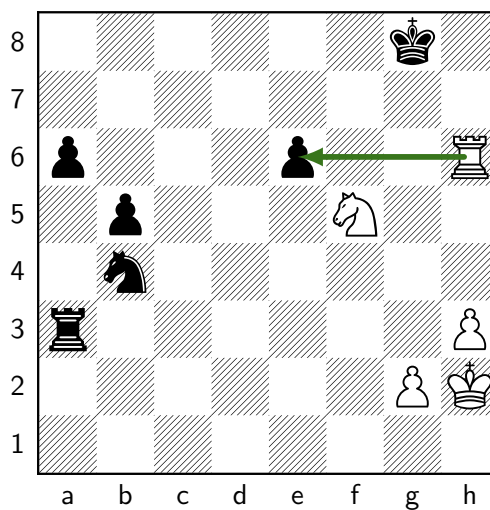


(d)

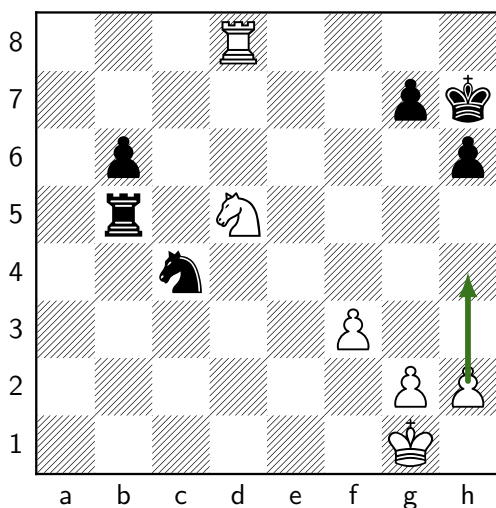
Figure 48: **Move 55, Black: Qc3+.** (a) Our position; Melanie plays Qc3+ (a3c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2423` (ply 98) [structural distance $d=46$, QAP obj=21]: the master played Bh2; under μ our Nd4 plays the role of that game's Bc7 (a **cross-type** correspondence); cross-type roles: our Qa3 as their Rc2, our Bb7 as their Pa5; transferred move d5c7. (c) Retrieved from `ccr1_top_p0002#2503` (ply 94) [structural distance $d=44$, QAP obj=20]: the master played Ra7+; under μ our Qa6 plays the role of that game's Ra8 (a **cross-type** correspondence); cross-type roles: our re2 as their Pb3, our Bb7 as their pe6; transferred move a3a4. (d) Retrieved from `ccr1_top_p0002#1492` (ply 101) [structural distance $d=49$, QAP obj=14]: the master played Kb5; under μ our Kg2 plays the role of that game's Kb3; cross-type roles: our Nd5 as their Ra4, our rd4 as their pc2; transferred move g7g8. Synthesis: no retrieved analog transferred to Qc3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



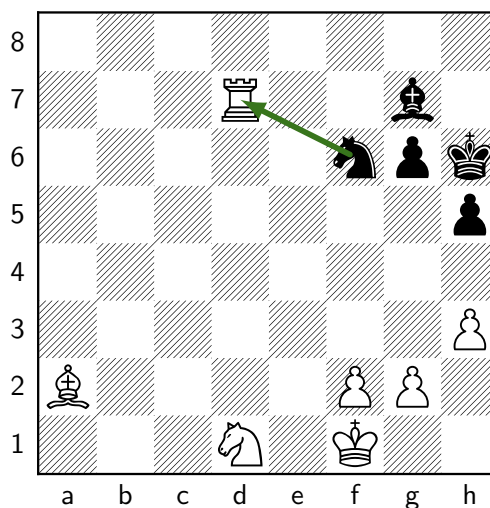
(a)



(b)

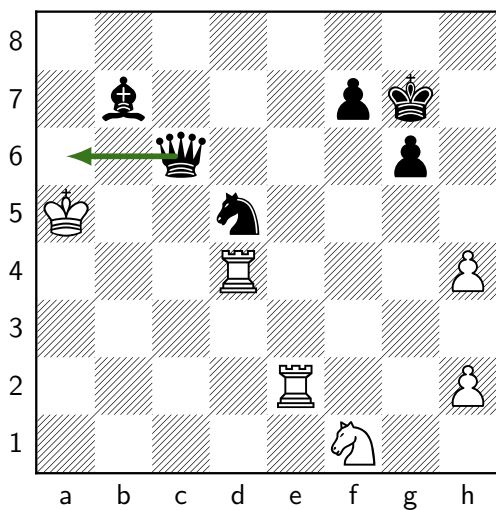


(c)

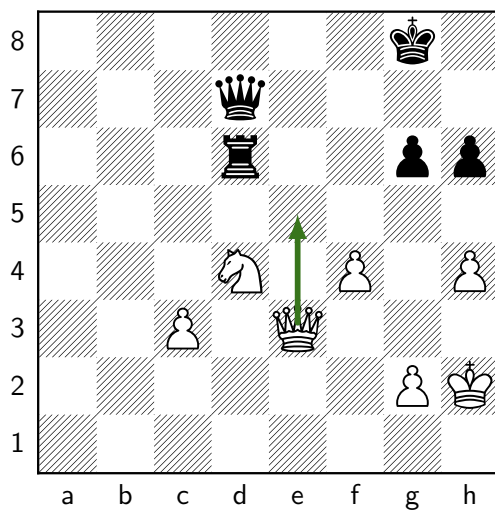


(d)

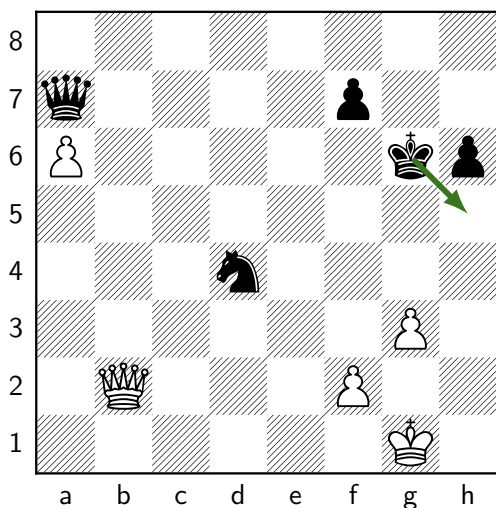
Figure 49: **Move 56, Black: Qc6+.** (a) Our position; Melanie plays Qc6+ (c3c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1993` (ply 106) [structural distance $d=48$, QAP obj=30]: the master played Rxe6; under μ our Qc6 plays the role of that game’s Rh6 (a **cross-type** correspondence); cross-type roles: our ph2 as their nb4, our nf1 as their pb5; transferred move c3d4. (c) Retrieved from `ccr1_top_p0000#3506` (ply 104) [structural distance $d=50$, QAP obj=19]: the master played h4; under μ our Pg3 plays the role of that game’s Ph2; cross-type roles: our re2 as their Pf3, our Bb7 as their nc4; transferred move g6g5. (d) Retrieved from `ccr1_top_p0000#481` (ply 95) [structural distance $d=50$, QAP obj=19]: the master played Nxd7; under μ our Qc6 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nd5 as their Bg7, our re2 as their ba2; transferred move c3c8. Synthesis: no retrieved analog transferred to Qc6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



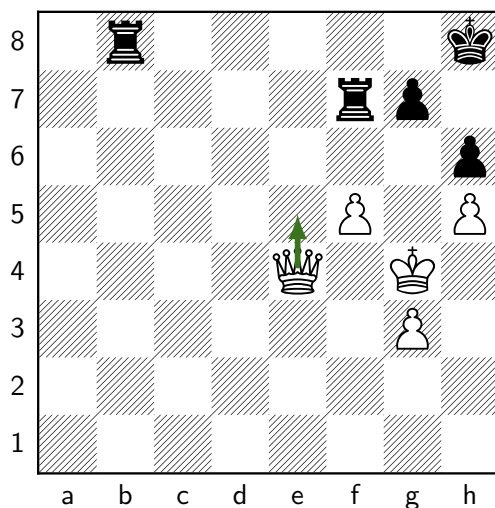
(a)



(b)



(c)



(d)

Figure 50: **Move 57, Black: Qa6#.** (a) Our position; Melanie plays Qa6# (c6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1058` (ply 114) [structural distance $d=57$, QAP obj=14]: the master played Qe5; under μ our Qc3 plays the role of that game’s Qe3; cross-type roles: our Bb7 as their Pc3, our nf1 as their ph6; transferred move c6c3. (c) Retrieved from `ccr1_top_p0001#2572` (ply 99) [structural distance $d=58$, QAP obj=11]: the master played Kh5; under μ our Kg2 plays the role of that game’s Kg3; cross-type roles: our Bb7 as their Pf7, our re2 as their pa6; transferred move g7g8. (d) Retrieved from `ccr1_top_p0000#3923` (ply 126) [structural distance $d=56$, QAP obj=3]: the master played Qe5; under μ our Qc3 plays the role of that game’s Qe4; cross-type roles: our Bb7 as their Pg3, our Nd5 as their Pf5; transferred move c6c5. Synthesis: no retrieved analog transferred to Qa6# – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.